

BEACON EDUCATIONAL SERIES

Integrated Science for Basic 7

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 7 (JHS 1) Students and Teachers

Beacon Educational Consult
2026 Edition

Integrated Science for Basic 7
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

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PREFACE

This book has been developed to support Junior High School Basic 7 (JHS 1) students and teachers in their study of Integrated Science, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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Paper 1: Agricultural

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A simple agricultural tool is one that is
 - A. made entirely of plastic
 - B. driven by a diesel engine
 - C. operated by hand and needs no engine
 - D. used only on very large farms
2. The tool used mainly for clearing weeds and cutting down bushes on a farm is the
 - A. rake
 - B. garden trowel
 - C. watering can
 - D. cutlass
3. The farm tool used for making ridges, mounds and weeding is the
 - A. wheelbarrow
 - B. hoe
 - C. secateurs
 - D. measuring tape
4. Seedlings in a nursery are watered gently with a
 - A. watering can
 - B. spraying machine for weedicide
 - C. shovel
 - D. bucket of ice
5. The tool used for gathering dry leaves and levelling the surface of a seedbed is the
 - A. knapsack sprayer
 - B. rake
 - C. axe
 - D. cutlass

- 6.** A tool used for digging holes for planting tree seedlings is the
- A. rake
 - B. pruning saw
 - C. watering can
 - D. pickaxe or mattock
- 7.** Secateurs are used on a farm mainly for
- A. pruning or cutting small branches of plants
 - B. carrying farm produce
 - C. watering young plants
 - D. ploughing the land
- 8.** Which of the following tools is used on an ANIMAL farm?
- A. A garden rake
 - B. A seed drill
 - C. A feeding trough brush
 - D. A pair of secateurs
- 9.** A wheelbarrow is used on a farm mainly for
- A. spraying insecticide
 - B. carrying manure, soil and harvested crops
 - C. cutting grass
 - D. measuring land
- 10.** Which of the following is used to protect the hands while working with farm tools?
- A. A watering can
 - B. Hand gloves
 - C. A cutlass sheath
 - D. A head pan
- 11.** The handling of an agricultural tool means the way the tool is
- A. shared among the farmers
 - B. painted at the factory
 - C. bought from the market
 - D. held and used during work
- 12.** The maintenance of agricultural tools means
- A. keeping the tools in a locked box for ever
 - B. buying new tools every season
 - C. selling old tools to other farmers
 - D. keeping the tools in good working condition

- 13.** Which of the following is NOT a reason for maintaining farm tools?
- A. To prevent accidents during work
 - B. To make the tools last longer
 - C. To make the tools heavier to use
 - D. To save the money spent on new tools
- 14.** Well-maintained farm tools help the farmer to
- A. harvest a poorer crop
 - B. spend more time on the farm
 - C. use more fuel
 - D. work faster and with less effort
- 15.** A blunt cutlass is dangerous to use because it
- A. becomes too light to hold
 - B. cuts the grass too quickly
 - C. may slip off the stem and injure the user
 - D. makes no sound when used
- 16.** The main reason for washing a hoe after use is to
- A. add weight to the blade
 - B. make the hoe look new for sale
 - C. soften the wooden handle
 - D. remove soil that would hold moisture and cause rusting
- 17.** Farm tools that are poorly maintained will MOST likely
- A. wear out quickly and need early replacement
 - B. become sharper on their own
 - C. work better with time
 - D. increase the farmer's harvest
- 18.** When walking with a cutlass on a farm, the blade should be
- A. held in front of the face
 - B. swung freely while walking
 - C. pointed downwards and away from the body
 - D. carried on the shoulder with the edge up
- 19.** A cutlass should be handed to another person by
- A. holding the blade and offering the handle first
 - B. offering the sharp edge first
 - C. dropping it at the person's feet
 - D. throwing it gently to the person

- 20.** Before using any farm tool, the user should FIRST
- A. check that the tool is in good condition and the handle is firm
 - B. paint the blade of the tool
 - C. sharpen the handle of the tool
 - D. wash the tool in the stream
- 21.** Farm tools should be used only for
- A. digging out stones with the blade of a cutlass
 - B. any work at all, to save time
 - C. games during break
 - D. the work for which they are made
- 22.** While weeding with a hoe, pupils should stand
- A. directly behind one another
 - B. very close together to finish quickly
 - C. a safe distance apart
 - D. in a single line facing each other
- 23.** The correct protective clothing for working with farm tools includes
- A. sandals and short sleeves
 - B. open slippers
 - C. a school uniform only
 - D. boots and hand gloves
- 24.** A tool with a loose handle should be
- A. repaired or the handle refitted before use
 - B. thrown into the bush
 - C. used quickly before it falls off
 - D. used only by the strongest pupil
- 25.** When a tool is not in use during work on the farm, it should be
- A. placed across the farm path
 - B. kept in a safe place with the sharp edge down or covered
 - C. hung on a growing plant
 - D. left lying in the grass with the edge up
- 26.** The maintenance practice that prevents the metal parts of farm tools from rusting is
- A. burying them in wet soil
 - B. leaving them in the sun for days
 - C. soaking them in water overnight
 - D. cleaning and oiling them after use

- 27.** A cutlass is kept in good cutting condition by
- A. rubbing it with wet sand only
 - B. sharpening it with a file or whetstone
 - C. leaving it in the rain to soften
 - D. heating it in a fire until it is red
- 28.** Wooden handles of farm tools are rubbed with oil or kept dry mainly to
- A. make them slippery to hold
 - B. make them heavier
 - C. change their colour
 - D. prevent cracking and attack by termites
- 29.** The BEST place to keep farm tools after work is
- A. in a pool of water to keep them cool
 - B. on the bare ground in the open
 - C. a dry, well-ventilated store, hung on racks
 - D. buried in the soil of the farm
- 30.** Rust on the blade of a hoe is BEST removed by
- A. painting over the rust
 - B. scraping or filing it off and then oiling the blade
 - C. heating the blade in a fire
 - D. soaking the blade in water
- 31.** Moving parts of a knapsack sprayer are kept working smoothly by
- A. tightening them until they cannot move
 - B. washing them with hot pepper water
 - C. oiling or greasing them regularly
 - D. leaving them in the sun
- 32.** A knapsack sprayer should be washed out thoroughly after use mainly because
- A. chemicals left inside will block the nozzle and corrode the parts
 - B. the chemicals will become more effective
 - C. washing makes the sprayer lighter to carry
 - D. the sprayer must be refilled with water for storage
- 33.** Which maintenance practice is MOST suitable for a wheelbarrow?
- A. Painting the wheel so that it cannot turn
 - B. Sharpening the tray with a file
 - C. Oiling the wheel and tightening the bolts
 - D. Soaking the whole barrow in water

- 34.** Tools are hung on racks or nails in the store rather than left on the floor so that they
- A. do not take up moisture from the floor and are not stepped on
 - B. will become lighter in weight
 - C. can be seen by visitors to the farm
 - D. can dry out and crack quickly
- 35.** During a visit to a farm in the community, the BEST way for pupils to record what they observe about tool maintenance is to
- A. argue with the farmer about his methods
 - B. take notes and write a report afterwards
 - C. remember everything without writing
 - D. borrow the farmer's tools and take them home
- 36.** Before pupils handle tools on a community farm, they should
- A. race to see who finishes first
 - B. be shown the correct way to use each tool by the teacher or farmer
 - C. remove their boots for comfort
 - D. share the tools among themselves and begin at once
- 37.** A set of written statements that tell workers how a tool should be used safely is called the
- A. harvest record of the farm
 - B. crop rotation plan
 - C. operational rules for handling the tool
 - D. price list of the tool
- 38.** A farmer in the community keeps his cutlass sharp, clean and hung in a dry store. The MOST likely result is that his cutlass will
- A. last for many years and work efficiently
 - B. rust within a few weeks
 - C. become blunt after one season
 - D. be rejected by other farmers
- 39.** A school farm club found that most tools in the store were rusty. The FIRST step they should take is to
- A. lock the store and stop farming
 - B. clean and oil the tools and repair the loose handles
 - C. throw all the tools away
 - D. buy a complete new set of tools

- 40.** The club then drew up a list of maintenance rules and put it on the store wall. This is useful mainly because it
- A. reminds every user of what to do after work with the tools
 - B. shows the price of each tool
 - C. decorates the wall of the store
 - D. stops pupils from entering the store

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 2: Animal Production

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An animal that is tamed and kept by people in the home or on a farm is called
 - A. a game animal
 - B. a wild animal
 - C. a domestic animal
 - D. a pest
2. Which of the following is NOT a domestic animal?
 - A. The pig
 - B. The antelope
 - C. The goat
 - D. The fowl
3. A group of animals of the same kind having the same features that they pass on to their young is called
 - A. a litter
 - B. a flock
 - C. a breed
 - D. a herd
4. Sanga, N'dama and White Fulani are breeds of
 - A. goats
 - B. cattle
 - C. sheep
 - D. fowls
5. West African Dwarf is a breed commonly used in Ghana for
 - A. goats
 - B. rabbits
 - C. turkeys
 - D. cattle only

- 6.** A farmer at Walewale says he keeps a 'flock'. The animals he is most likely talking about are
- A. pigs
 - B. rabbits
 - C. sheep or fowls
 - D. cattle
- 7.** An animal with a stomach divided into four chambers is described as
- A. a monogastric animal
 - B. a scavenger
 - C. a carnivore
 - D. a ruminant
- 8.** Which of the following animals is a ruminant?
- A. Fowl
 - B. Sheep
 - C. Pig
 - D. Rabbit
- 9.** The habit of bringing swallowed food back to the mouth to chew it again is called
- A. scavenging
 - B. grazing
 - C. browsing
 - D. chewing the cud
- 10.** Ruminants are able to feed mainly on grass because
- A. grass contains no fibre at all
 - B. they swallow stones to grind the grass
 - C. the micro-organisms in their stomach chambers break down the grass
 - D. they have very sharp cutting teeth in the upper jaw
- 11.** Which of the following groups contains ONLY ruminants?
- A. Cattle, sheep and goats
 - B. Sheep, fowls and goats
 - C. Cattle, pigs and goats
 - D. Goats, rabbits and sheep
- 12.** A goat and a pig are given the same heap of fresh grass. The goat does better on it because the goat
- A. eats much faster than the pig
 - B. can digest the fibre in the grass with the help of its stomach micro-organisms
 - C. has a longer mouth than the pig
 - D. does not need any other feed at all

- 13.** The four chambers of the stomach of a ruminant include the
- A. gizzard
 - B. crop
 - C. rumen
 - D. caecum only
- 14.** An animal with a simple stomach of one chamber is described as
- A. a cud chewer
 - B. a monogastric animal
 - C. a ruminant
 - D. a herbivore
- 15.** Which of the following is a monogastric animal?
- A. The sheep
 - B. The cow
 - C. The pig
 - D. The goat
- 16.** Birds kept on a farm for their meat and eggs are known as
- A. draught animals
 - B. livestock feed
 - C. ruminants
 - D. poultry
- 17.** Which of the following is NOT poultry?
- A. The rabbit
 - B. The turkey
 - C. The guinea fowl
 - D. The duck
- 18.** The part of the digestive system of a fowl that grinds its food is the
- A. gizzard
 - B. rumen
 - C. crop only
 - D. udder
- 19.** A pig is usually fed on kitchen waste, maize bran and other concentrates rather than on grass alone because the pig
- A. refuses to eat any plant material
 - B. has a simple stomach and cannot digest much fibre
 - C. is too large to walk to the grass
 - D. has no teeth for cutting grass

- 20.** A rabbit is a monogastric animal, yet it feeds mainly on leaves and grass. This shows that
- A. leaves contain no fibre
 - B. the rabbit is a ruminant
 - C. all monogastric animals eat only meat
 - D. monogastric animals can also be herbivores
- 21.** Ruminants, monogastric animals and poultry are alike in that all of them
- A. have four stomach chambers
 - B. lay eggs
 - C. are kept only for their milk
 - D. feed, breathe, grow and produce young ones
- 22.** Which of the following is a difference between a fowl and a goat?
- A. The goat lays eggs while the fowl does not
 - B. The fowl has feathers and two legs while the goat has hair and four legs
 - C. The fowl is a ruminant while the goat is not
 - D. The goat has a gizzard while the fowl has a rumen
- 23.** A pupil wrote: 'Cattle and pigs are both ruminants because both are kept on farms.' The statement is wrong because
- A. only wild animals are ruminants
 - B. pigs have four stomach chambers
 - C. being kept on a farm does not determine the type of stomach an animal has
 - D. cattle are not kept on farms
- 24.** A farmer who has very little money for feed but plenty of grass on his land should best keep
- A. broiler fowls only
 - B. rabbits only
 - C. pigs only
 - D. sheep and goats
- 25.** Which observation would BEST help a pupil to decide whether an animal is a ruminant?
- A. Watching whether the animal brings food back to its mouth to chew again while resting
 - B. Noting the colour of its hair
 - C. Measuring the length of its tail
 - D. Counting the number of its legs

- 26.** Poultry are kept in large numbers on a small piece of land more easily than cattle mainly because poultry
- A. produce no waste at all
 - B. are small, grow quickly and need less space
 - C. cannot move about
 - D. do not need any feed
- 27.** The use of an animal to meet the needs of the family at home is called
- A. industrial use
 - B. wild use
 - C. commercial use
 - D. domestic use
- 28.** Which of the following is a domestic use of a fowl?
- A. Providing eggs and meat for the family
 - B. Exporting frozen chicken
 - C. Supplying a hotel with eggs every week
 - D. Selling day-old chicks in the market
- 29.** A farmer uses his bull to pull a plough on his farm. This use of the animal is called
- A. draught (work) use
 - B. commercial sale
 - C. a breed
 - D. a by-product
- 30.** Cattle dung spread on a food crop farm is useful because it
- A. kills all insects on the farm
 - B. prevents rain from falling
 - C. makes the soil hard and dry
 - D. adds nutrients and humus to the soil
- 31.** A family keeps goats and sells two of them to pay school fees. The goats have been used
- A. as draught animals
 - B. commercially
 - C. domestically only
 - D. as poultry
- 32.** Which of the following best shows the domestic use of a cow in a home?
- A. Renting the cow to another farmer
 - B. Supplying milk to a factory
 - C. Taking milk from the cow for the family to drink
 - D. Selling the cow at the market

- 33.** Animals are also kept for social and cultural purposes in Ghana, such as
- A. providing sheep and fowls for naming ceremonies and festivals
 - B. guarding the national borders
 - C. producing electricity for the home
 - D. pulling vehicles in the city
- 34.** Keeping animals mainly in order to sell them or their products for money is called
- A. cultural use
 - B. domestic use
 - C. subsistence use
 - D. commercial use
- 35.** A useful material obtained from an animal apart from its meat is called
- A. a by-product
 - B. a carcass only
 - C. a ration
 - D. a breed
- 36.** Which of the following is an animal by-product?
- A. Maize bran fed to the animal
 - B. Water given to the animal
 - C. Grass grown on the farm
 - D. Hide used for making leather
- 37.** The skin of a goat is used in many Ghanaian communities to
- A. feed other animals
 - B. replace roofing sheets
 - C. make the heads of drums and leather goods
 - D. build the walls of houses
- 38.** Animal droppings from a poultry house are best described as
- A. a by-product that can be used as manure
 - B. a type of animal feed for cattle
 - C. a disease of poultry
 - D. a waste of no use at all
- 39.** A farmer near Kumasi keeps 500 layers and sells the eggs to shops in the town every week. The main purpose of his farm is
- A. cultural
 - B. experimental
 - C. commercial
 - D. domestic

- 40.** A young farmer has a small plot behind her house and wants a quick return on the little money she has. The BEST advice is to
- A. start with a herd of cattle, because cattle are the largest animals
 - B. start with a small flock of fowls, because they grow fast and produce eggs and meat within a short time
 - C. keep one pig and wait for it to produce manure only
 - D. keep no animals until she can afford a large farm

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 3: Climate Change and C

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A sustainable energy source is one that
 - A. is found only outside Ghana
 - B. is always the cheapest to buy
 - C. can be used again and again without being used up
 - D. burns with the brightest flame
2. Which of the following is a renewable source of energy?
 - A. petrol
 - B. sunlight
 - C. coal
 - D. diesel
3. The energy of moving air is called
 - A. tidal energy
 - B. wind energy
 - C. chemical energy
 - D. nuclear energy
4. Energy obtained from the heat and light of the Sun is
 - A. geothermal energy
 - B. sound energy
 - C. solar energy
 - D. fossil energy
5. Electricity produced at the Akosombo Dam is obtained from
 - A. burning petrol
 - B. burning charcoal
 - C. the wind
 - D. falling water
6. Which of the following is NOT a fossil fuel?
 - A. coal
 - B. natural gas
 - C. firewood
 - D. crude oil

- 7.** Fossil fuels are described as non-renewable because they
- A. cannot burn in air
 - B. take millions of years to form and are being used up quickly
 - C. are found in only one country
 - D. are made in factories every day
- 8.** The gas produced when organic waste rots in a closed tank without air is
- A. biogas
 - B. oxygen
 - C. nitrogen
 - D. chlorine
- 9.** Planting trees on bare land to protect the environment is called
- A. afforestation
 - B. urbanisation
 - C. deforestation
 - D. desertification
- 10.** A pupil who switches off the lights and the fan before leaving the classroom is practising
- A. energy wastage
 - B. energy production
 - C. energy conservation
 - D. energy transformation
- 11.** Replacing filament bulbs with LED bulbs in a school helps the environment MAINLY because LED bulbs
- A. are made of glass only
 - B. are brighter than the Sun
 - C. never need to be switched off
 - D. use less electricity for the same amount of light
- 12.** Which of these practices adds the LEAST pollution to the air?
- A. drying clothes in the Sun
 - B. running a petrol generator all day
 - C. burning rubbish in the yard
 - D. burning old tyres
- 13.** The use of an improved cooking stove that needs less firewood helps the environment because it
- A. costs more money to buy
 - B. increases the heat lost to the air
 - C. makes the food taste better
 - D. reduces the number of trees cut down and the smoke produced

- 14.** Walking or cycling to school instead of being driven helps the environment because it
- A. reduces the exhaust gases released into the air
 - B. increases the number of vehicles on the road
 - C. makes the roads wider
 - D. uses more fuel
- 15.** Which of the following BEST describes a green economy?
- A. an economy based only on farming
 - B. an economy that creates wealth while protecting the environment
 - C. an economy that uses no energy at all
 - D. an economy in which only green goods are sold
- 16.** Sorting household waste so that bottles and tins are collected separately supports
- A. recycling
 - B. open burning
 - C. littering
 - D. land filling of everything
- 17.** The greenhouse effect is the
- A. reflection of all sunlight back into space
 - B. growing of green plants inside a glass house
 - C. cooling of the Earth by rain clouds
 - D. trapping of heat near the Earth's surface by gases in the atmosphere
- 18.** Which of the following is a greenhouse gas?
- A. helium
 - B. nitrogen
 - C. oxygen
 - D. carbon dioxide
- 19.** Besides carbon dioxide, another important greenhouse gas is
- A. hydrogen
 - B. methane
 - C. neon
 - D. argon
- 20.** Carbon dioxide is added to the atmosphere MAINLY by
- A. the burning of fuels such as petrol, coal and firewood
 - B. the freezing of water
 - C. the evaporation of sea water
 - D. the rusting of iron

- 21.** Green plants help to reduce the amount of carbon dioxide in the air because they
- A. take in carbon dioxide during photosynthesis
 - B. give out carbon dioxide at night only
 - C. turn carbon dioxide into soil
 - D. store carbon dioxide in their roots as salt
- 22.** A steady rise in the average temperature of the Earth is called
- A. condensation
 - B. global dimming
 - C. evaporation
 - D. global warming
- 23.** Which of the following is a likely effect of global warming?
- A. a permanent end to the rainy season everywhere
 - B. melting of ice at the poles and a rise in sea level
 - C. an increase in the number of planets
 - D. a fall in the amount of carbon dioxide in the air
- 24.** Cutting down large areas of forest increases the greenhouse effect because it
- A. cools the soil too quickly
 - B. adds oxygen to the air
 - C. stops the wind from blowing
 - D. removes the trees that would have taken in carbon dioxide
- 25.** The layer of gases that surrounds the Earth and holds in some of its heat is the
- A. biosphere
 - B. atmosphere
 - C. lithosphere
 - D. hydrosphere
- 26.** Climate change is BEST described as
- A. the change of weather from morning to afternoon
 - B. a long-term change in the usual pattern of weather of a place
 - C. the movement of clouds across the sky
 - D. the change of seasons within a single week
- 27.** Which of the following is an effect of climate change on farming in Ghana?
- A. crops no longer need water
 - B. all soils become more fertile
 - C. rains become unreliable, so planting times are harder to predict
 - D. weeds stop growing on farms

- 28.** Flooding in coastal towns is made worse by climate change MAINLY because
- A. rivers flow backwards during the night
 - B. the Moon moves nearer to the Earth
 - C. the level of the sea rises and storms become stronger
 - D. the soil becomes lighter in colour
- 29.** A community that wants to reduce the greenhouse gases it produces should
- A. cut down trees to make more charcoal
 - B. use solar lamps and plant trees instead of burning more fuel
 - C. burn its rubbish in open heaps every evening
 - D. buy more petrol generators for every house
- 30.** The BEST reason for using public transport, or sharing a vehicle, is that it
- A. removes the need for roads
 - B. makes journeys take longer
 - C. carries many people while releasing less exhaust gas per person
 - D. allows drivers to use more fuel
- 31.** Burning farm residues in the open field after harvest is discouraged because it
- A. increases the number of earthworms
 - B. adds water to the soil
 - C. releases carbon dioxide and smoke and destroys soil organisms
 - D. makes the field cooler
- 32.** Which pair of actions would MOST reduce a school's contribution to climate change?
- A. switching off unused appliances and planting trees in the compound
 - B. clearing the school garden and paving all the ground
 - C. burning waste paper daily and using more generators
 - D. leaving taps running and buying more bulbs
- 33.** A country is said to use clean energy when its electricity comes mainly from
- A. firewood and charcoal
 - B. old tyres and plastics
 - C. coal and crude oil
 - D. the Sun, wind and water
- 34.** A simple solar dryer used to dry pepper and fish works by
- A. burning charcoal under the trays
 - B. trapping the Sun's heat so that the warm air removes moisture
 - C. adding salt to the food in a tank
 - D. cooling the food with ice

- 35.** In a biogas project, the material fed into the digester is
- A. animal dung and other organic waste
 - B. plastic bags and metal
 - C. sand and gravel
 - D. broken bottles and tins
- 36.** Apart from gas for cooking, a biogas plant also provides
- A. slurry that can be used as manure on the farm
 - B. pure drinking water
 - C. cement for building
 - D. petrol for vehicles
- 37.** A school wishing to light its classrooms during a power cut, without polluting the air, should install
- A. a petrol generator
 - B. a diesel generator
 - C. a charcoal stove
 - D. solar panels with a storage battery
- 38.** Before building a small energy project, the FIRST step a class should take is to
- A. paint the school wall
 - B. call the whole town for a party
 - C. plan and find out which source of energy is available in the area
 - D. buy all the materials at once
- 39.** Which of the following would BEST show that a class project on sustainable energy has succeeded?
- A. the device uses more fuel than before
 - B. the device is the most expensive in the district
 - C. the device works and reduces the fuel or electricity the school uses
 - D. the device is kept in a locked store
- 40.** Green jobs are jobs that
- A. require workers to wear green uniforms
 - B. are done only on farms
 - C. are done only during the rainy season
 - D. help to protect the environment while providing a living

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 4: Conversion and Conservation

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The law of conservation of energy states that energy can be
 - A. created whenever it is needed
 - B. destroyed after it has been used
 - C. stored only in dry cells
 - D. changed from one form to another but cannot be created or destroyed
2. The unit in which energy is measured is the
 - A. joule
 - B. newton
 - C. ampere
 - D. volt
3. A system into which no energy enters and from which no energy escapes is described as a
 - A. moving system
 - B. closed system
 - C. heated system
 - D. wasted system
4. In a closed system, the total amount of energy at the end of a change is
 - A. always less than at the beginning
 - B. the same as the total amount at the beginning
 - C. equal to zero
 - D. always more than at the beginning
5. A dry cell supplies 60 J of energy to a torch bulb. The bulb gives out 18 J as light. The energy given out as heat is
 - A. 60 J
 - B. 78 J
 - C. 18 J
 - D. 42 J

- 6.** Which of the following statements about energy changes is TRUE?
- A. Energy is only changed from one useful form into other forms
 - B. Energy disappears when a machine stops working
 - C. New energy is made inside a dry cell
 - D. Energy can be weighed on a balance
- 7.** A pupil said that the electricity used by a fan is 'lost' when the fan is switched off. The correct statement is that the energy has been
- A. changed into kinetic energy, sound and heat
 - B. destroyed by the motor
 - C. stored inside the blades
 - D. returned unchanged to the socket
- 8.** A machine is said to waste energy when some of the energy supplied to it is
- A. destroyed completely inside the machine
 - B. changed into forms that are not useful for the job
 - C. weighed and thrown away
 - D. returned to the source of the energy
- 9.** The energy change that takes place in a dry-cell torch when it is switched on is
- A. electrical to chemical to sound
 - B. light to electrical to chemical
 - C. chemical to electrical to light and heat
 - D. heat to chemical to light
- 10.** A solar panel on the roof of a clinic changes
- A. solar energy into electrical energy
 - B. sound energy into electrical energy
 - C. electrical energy into solar energy
 - D. heat energy into chemical energy
- 11.** An electric pressing iron changes electrical energy mainly into
- A. heat energy
 - B. chemical energy
 - C. sound energy
 - D. light energy
- 12.** The main energy change in a loudspeaker of a radio set is
- A. chemical energy to light energy
 - B. sound energy to electrical energy
 - C. heat energy to kinetic energy
 - D. electrical energy to sound energy

- 13.** An electric fan changes electrical energy mainly into
- A. kinetic energy
 - B. chemical energy
 - C. light energy
 - D. nuclear energy
- 14.** A microphone used at a school gathering changes
- A. heat energy into sound energy
 - B. sound energy into electrical energy
 - C. electrical energy into sound energy
 - D. light energy into sound energy
- 15.** When a mobile phone is being charged, the energy change is
- A. sound energy to heat energy
 - B. light energy to electrical energy
 - C. chemical energy to electrical energy
 - D. electrical energy to chemical energy
- 16.** Green plants carry out an energy conversion in which
- A. chemical energy is changed into light energy
 - B. heat energy is changed into sound energy
 - C. light energy is changed into chemical energy in food
 - D. kinetic energy is changed into light energy
- 17.** Which device changes kinetic energy into electrical energy?
- A. A generator
 - B. An electric motor
 - C. A pressing iron
 - D. A dry cell
- 18.** In a hydroelectric station such as Akosombo, the energy of the stored water is changed in the order
- A. chemical to heat to electrical
 - B. kinetic to chemical to electrical
 - C. potential to kinetic to electrical
 - D. electrical to kinetic to potential
- 19.** A boy winds up a toy car and releases it. The energy change is
- A. heat energy to potential energy
 - B. sound energy to kinetic energy
 - C. kinetic energy to chemical energy
 - D. elastic potential energy to kinetic energy

- 20.** A candle burning in a dark room changes chemical energy into
- A. electrical and nuclear energy
 - B. potential and elastic energy
 - C. light and heat energy
 - D. sound and kinetic energy
- 21.** A motor is supplied with 100 J of electrical energy. It gives 70 J of kinetic energy. The energy that appears as heat and sound is
- A. 100 J
 - B. 30 J
 - C. 70 J
 - D. 170 J
- 22.** A lamp supplied with 50 J of electrical energy gives out 10 J of light. The fraction of the energy that is useful is
- A. the whole
 - B. four-fifths
 - C. one-half
 - D. one-fifth
- 23.** In every energy conversion, some of the energy is usually changed into
- A. solar energy
 - B. nuclear energy
 - C. chemical energy
 - D. heat energy
- 24.** A pupil lists the energy chain of a petrol lorry as: chemical to to kinetic. The missing form is
- A. light energy
 - B. nuclear energy
 - C. heat energy
 - D. sound energy
- 25.** Which of the following energy chains is CORRECT for an electric kettle boiling water?
- A. kinetic energy to light energy
 - B. heat energy to electrical energy
 - C. electrical energy to heat energy
 - D. chemical energy to sound energy
- 26.** A filament bulb gives out far more heat than light. This means that most of the energy supplied to it is
- A. destroyed
 - B. stored
 - C. returned to the mains
 - D. wasted

- 27.** An LED lamp is preferred to a filament bulb for lighting a classroom because the LED lamp
- A. gives out more heat
 - B. uses a larger current
 - C. changes a greater part of the electrical energy into light
 - D. needs no switch to control it
- 28.** Oiling the moving parts of a machine saves energy because it reduces the energy wasted as
- A. light given out by the metal
 - B. heat and sound caused by friction
 - C. electrical energy in the wires
 - D. chemical energy in the oil
- 29.** Of the energy supplied to it, the useful energy given out by a machine is always
- A. equal to the energy supplied
 - B. less than the energy supplied
 - C. equal to zero
 - D. more than the energy supplied
- 30.** Heat lost through the walls of an electric oven is an example of
- A. destroyed energy
 - B. created energy
 - C. wasted energy
 - D. stored energy
- 31.** A pupil claimed to have built a machine that works for ever without any supply of energy. The claim must be false because
- A. machines are always made of metal
 - B. no machine can give out heat
 - C. energy cannot be created; a machine must be supplied with energy
 - D. every machine must have an electric motor
- 32.** Energy conservation means
- A. using energy carefully so that less of it is wasted
 - B. making new energy in a laboratory
 - C. destroying energy that is not needed
 - D. using as much energy as possible
- 33.** Which of the following is the BEST way of conserving electrical energy at home?
- A. Using filament bulbs in every room
 - B. Switching off lights and fans in rooms that are not in use
 - C. Leaving the television on all night

- D. Opening the fridge door often
- 34.** Drying clothes in the open air instead of using a drying machine conserves energy because it
- A. increases the cost of electricity
 - B. uses energy from the Sun, which is free and renewable
 - C. destroys the electrical energy saved
 - D. makes the clothes heavier
- 35.** Ironing all the family's clothes at one time rather than one garment at a time conserves energy because
- A. the iron gives out no heat
 - B. the clothes need less water
 - C. the iron is heated once instead of many times
 - D. the iron works at a lower voltage
- 36.** A sign of energy being wasted in a school is
- A. an LED lamp used in the library
 - B. a kettle switched off after the water has boiled
 - C. a classroom with the windows opened for air
 - D. a classroom with the lights and fans left on at break time
- 37.** Which of the following sources of energy will NOT run out if it is used wisely?
- A. Sunlight
 - B. Crude oil
 - C. Coal
 - D. Natural gas
- 38.** A reason for conserving energy in Ghana is that it
- A. increases the number of power cuts
 - B. raises the cost of electricity for everybody
 - C. makes dry cells last a shorter time
 - D. reduces the fuel burnt and so protects the environment
- 39.** Putting a lid on a cooking pot conserves energy because it
- A. changes the heat into light
 - B. keeps the heat in the pot so that the food cooks faster
 - C. adds more heat to the food
 - D. makes the fire burn more fuel
- 40.** A school wishes to teach the community about saving energy. The MOST suitable activity is to
- A. make posters and leaflets on energy saving and display them in the town
 - B. ask every home to stop using electricity
 - C. switch off the school water pump for a term

D. sell dry cells to the community

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps in any calculation, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram, an energy-flow diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 5: Crop Production

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Substances that plants take in from the soil for healthy growth are called
 - A. plant hormones
 - B. plant nutrients
 - C. plant cells
 - D. plant fibres
2. The part of the plant through which most nutrients are taken in from the soil is the
 - A. root
 - B. flower
 - C. fruit
 - D. leaf
3. Plant nutrients are taken up by the roots
 - A. as dry solid grains
 - B. only when the soil is completely dry
 - C. as a gas from the air
 - D. dissolved in soil water
4. Any substance added to the soil to supply nutrients to crops is known as
 - A. a fertilizer
 - B. a fungicide
 - C. a herbicide
 - D. a pesticide
5. Maize plants on a school farm are short, with pale yellow leaves, while those on the next plot are tall and dark green. The most likely cause is that the soil of the first plot
 - A. lacks enough plant nutrients
 - B. is in a cooler part of the school
 - C. has too many stones
 - D. contains too much air

- 6.** Which of the following is the chief reason for adding plant nutrients to a school garden year after year?
- A. Plants cannot take in water without fertilizer
 - B. The crops harvested each year remove nutrients from the soil
 - C. Fertilizers make the soil heavier
 - D. Nutrients make the soil change colour
- 7.** A source of plant nutrients obtained from the remains of plants and animals is described as
- A. mineral
 - B. artificial
 - C. organic
 - D. inorganic
- 8.** Which of the following is an organic source of plant nutrients?
- A. Compost
 - B. Urea
 - C. Ammonium sulphate
 - D. Muriate of potash
- 9.** Rotted plant remains, kitchen waste and animal dung heaped together and allowed to decay form
- A. gravel
 - B. cement
 - C. charcoal
 - D. compost
- 10.** Dung and urine of farm animals mixed with their bedding is known as
- A. wood ash
 - B. farmyard manure
 - C. green manure
 - D. liquid fertilizer
- 11.** A farmer ploughs a young leguminous crop into the soil while it is still fresh and green. This practice provides
- A. green manure
 - B. inorganic fertilizer
 - C. poultry litter only
 - D. farmyard manure
- 12.** Which of the following organic materials is likely to be collected from a poultry house?
- A. Granules of NPK fertilizer
 - B. Bags of lime
 - C. Droppings mixed with wood shavings
 - D. Crystals of ammonium sulphate

- 13.** Organic sources of plant nutrients improve the soil in a way that inorganic fertilizers do not, because they
- A. supply nutrients instantly to the crop
 - B. make the soil completely free of germs
 - C. add humus, which improves the structure of the soil and its ability to hold water
 - D. stop all weeds from growing
- 14.** A source of plant nutrients that is manufactured in a factory from mineral substances is described as
- A. natural
 - B. decayed
 - C. organic
 - D. inorganic
- 15.** Which of the following is an inorganic fertilizer?
- A. Compost
 - B. NPK 15-15-15
 - C. Green manure
 - D. Poultry manure
- 16.** The letters N, P and K on a bag of fertilizer stand for
- A. nitrogen, phosphorus and potassium
 - B. nickel, phosphorus and potassium
 - C. nitrogen, phosphorus and carbon
 - D. nitrogen, potassium and calcium
- 17.** Which of the following is supplied mainly by sulphate of ammonia and urea?
- A. Calcium
 - B. Phosphorus
 - C. Nitrogen
 - D. Potassium
- 18.** Inorganic fertilizers act faster on a crop than organic manures because they
- A. are always applied in larger quantities
 - B. are heavier than manure
 - C. contain living organisms
 - D. dissolve quickly in soil water and are taken up at once
- 19.** A danger of using too much inorganic fertilizer on a bed of vegetables is that it may
- A. scorch (burn) the plants and harm the soil
 - B. prevent rain from falling on the farm
 - C. turn the soil into stone
 - D. make the plants grow too slowly to harvest

- 20.** Which of the following statements about inorganic fertilizers is correct?
- A. They are always cheaper than organic manures
 - B. They supply nutrients in a concentrated form, so only small quantities are needed
 - C. They are made by heaping plant remains to decay
 - D. They add humus to the soil
- 21.** Which of the following pairs contains ONLY organic sources of plant nutrients?
- A. Compost and farmyard manure
 - B. Urea and poultry manure
 - C. Superphosphate and urea
 - D. NPK and green manure
- 22.** Compared with inorganic fertilizers, organic manures are
- A. quicker in their action on the crop
 - B. light, so very small quantities are enough
 - C. bulky, so large quantities must be applied
 - D. always sold in sealed sacks
- 23.** A farmer needs 2 kg of NPK fertilizer to supply the nutrients for one bed of vegetables. To supply the same nutrients with poultry manure he would need
- A. exactly 2 kg as well
 - B. less than 1 kg
 - C. no manure at all, since manure has no nutrients
 - D. a much larger quantity, measured in many kilograms
- 24.** One advantage of organic manure over inorganic fertilizer is that organic manure
- A. is sold in small light bags
 - B. acts within one or two days
 - C. contains only one nutrient
 - D. releases its nutrients slowly over a long period
- 25.** A school has little money but keeps goats and sweeps up plenty of leaves and kitchen waste. The best source of nutrients for its garden is
- A. no nutrients at all until money is available
 - B. bags of imported NPK fertilizer
 - C. compost and goat manure made on the school compound
 - D. liquid fertilizer bought from the city
- 26.** A farmer who uses only inorganic fertilizer year after year without adding any organic matter will most likely find that
- A. the soil becomes richer in humus each year
 - B. the soil loses its structure and holds less water
 - C. weeds disappear from the farm completely

D. the crop no longer needs water

27. Well-made compost is usually

- A. clear, colourless and watery
- B. yellow, dry and powdery
- C. white, hard and shiny
- D. dark brown to black, crumbly and earthy-smelling

28. NPK fertilizer is commonly supplied as

- A. a dark crumbly heap
- B. a wet, smelly paste
- C. small hard granules in a sealed sack
- D. a block of pressed leaves

29. Which of the following nutrient sources is a liquid?

- A. Farmyard manure
- B. Liquid fertilizer mixed with water for spraying
- C. Dry poultry litter
- D. Muriate of potash crystals

30. Fresh poultry droppings differ from dried poultry manure in that the fresh droppings are

- A. dry, light and odourless
- B. hard, white crystals
- C. clear and colourless
- D. wet, sticky and strong-smelling

31. Inorganic fertilizer is kept in a sealed bag in a dry store mainly because it

- A. turns into compost when it is left open
- B. is food for farm animals
- C. absorbs moisture from the air and cakes into lumps
- D. gives off a pleasant smell

32. Why is a bulky organic manure usually worked into the soil rather than left on the surface?

- A. So that it can be seen by the farmer
- B. So that the soil can become harder
- C. So that it decays in the soil and releases its nutrients where the roots are
- D. So that birds can feed on it easily

33. The physical form of a nutrient source affects how it is applied. A fine powder is harder to spread by hand on a windy day than granules because the powder

- A. dissolves in the air
- B. has no nutrients in it

- C. is heavier than the granules
 - D. is light and is blown away easily
- 34.** Spreading fertilizer evenly over the whole surface of a bed is called
- A. broadcasting
 - B. ring application
 - C. foliar spraying
 - D. drilling
- 35.** Placing fertilizer in a circle round a plant, a short distance from the stem, is known as
- A. top dressing of the leaves
 - B. ring application
 - C. broadcasting
 - D. deep ploughing
- 36.** Fertilizer is placed a short distance away from the stem of a plant rather than touching it in order to
- A. allow the wind to blow it away
 - B. avoid scorching the stem and roots while still feeding the plant
 - C. make the plant grow in a circle
 - D. keep the fertilizer away from the roots
- 37.** Liquid fertilizer that is sprayed onto the leaves of a crop is said to be applied as
- A. a foliar spray
 - B. a basal dressing
 - C. a ring application
 - D. a compost heap
- 38.** Compost is best applied to a vegetable bed
- A. sprinkled on the leaves of the growing crop
 - B. during land preparation, worked into the soil before planting
 - C. on the day of harvesting
 - D. only after the crop has been uprooted
- 39.** A pupil applies fertilizer to a bed in the middle of a hot, dry afternoon and the crop later shows burnt leaf edges. The best advice to the pupil is to
- A. apply twice as much fertilizer next time
 - B. stop using any fertilizer on the bed again
 - C. apply fertilizer when the soil is moist, such as after a light rain or watering
 - D. apply the fertilizer directly on the leaves instead

- 40.** A school garden has one wheelbarrow of poultry manure and one small bag of NPK for four vegetable beds. The BEST plan is to
- A. spray both the manure and the NPK onto the leaves of the crop
 - B. mix the manure and the NPK together and heap them in one corner
 - C. pour all the NPK onto one bed and leave the manure unused
 - D. work the manure into all four beds during land preparation and side-dress the growing crop with small measured amounts of NPK

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 6: Ecosystems

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An ecosystem is best described as
 - A. a place where only animals are found
 - B. a group of animals of the same kind
 - C. a unit in which living things and their non-living surroundings interact
 - D. the soil in which plants grow
2. An ecosystem is said to be self-sustaining because
 - A. no living thing ever dies in it
 - B. it never changes from year to year
 - C. its parts supply one another with what they need to continue
 - D. it is kept going by human beings
3. Which of the following is an example of an ecosystem?
 - A. A piece of iron sheet
 - B. A pond
 - C. A bucket of sand
 - D. A single mango fruit
4. The place where a plant or an animal lives in an ecosystem is called its
 - A. nutrient
 - B. community
 - C. population
 - D. habitat
5. All the living things of different kinds living together in one place form
 - A. a community
 - B. a habitat
 - C. an organ
 - D. a population
6. Which of the following best explains why the Owabi forest can be called an ecosystem?
 - A. It is found in the Ashanti Region
 - B. It contains very many trees
 - C. Its plants, animals, soil, water and air act on one another as one unit

- D. It is protected by the government
- 7.** The study of ecosystems is important mainly because it
- A. helps us to use and protect our surroundings wisely
 - B. shows us how to build better houses
 - C. helps us to kill dangerous animals
 - D. teaches us the names of all plants
- 8.** An ecosystem found on land is described as
- A. artificial
 - B. terrestrial
 - C. aquatic
 - D. arboreal
- 9.** An ecosystem found in water is described as
- A. arid
 - B. terrestrial
 - C. arboreal
 - D. aquatic
- 10.** An ecosystem made up of the life found on trees is described as
- A. terrestrial
 - B. arboreal
 - C. marine
 - D. aquatic
- 11.** Which of the following is an aquatic ecosystem?
- A. The Kakum forest
 - B. The Volta Lake
 - C. The Accra plains
 - D. A baobab tree
- 12.** A grassland in the Northern Region of Ghana is an example of
- A. a terrestrial ecosystem
 - B. a man-made lake
 - C. an arboreal ecosystem
 - D. an aquatic ecosystem
- 13.** Monkeys, birds, orchids and insects living in the branches of a tall forest tree make up
- A. a desert ecosystem
 - B. an arboreal ecosystem
 - C. an aquatic ecosystem
 - D. a soil ecosystem

- 14.** Which of the following pairs are BOTH aquatic ecosystems?
- A. The sea and a grassland
 - B. A mangrove tree and a rock
 - C. A pond and a forest
 - D. A river and the sea
- 15.** The living parts of an ecosystem are called
- A. mineral components
 - B. physical components
 - C. abiotic components
 - D. biotic components
- 16.** Which of the following is a biotic component of a pond?
- A. Sunlight
 - B. Tilapia
 - C. Mud
 - D. Water
- 17.** Green plants in an ecosystem are called producers because they
- A. produce the water of the ecosystem
 - B. break down dead bodies
 - C. are eaten by all other organisms
 - D. make their own food using sunlight
- 18.** Animals in an ecosystem are called consumers because they
- A. produce oxygen for other organisms
 - B. use sunlight to make food
 - C. depend on other organisms for their food
 - D. live only in water
- 19.** Bacteria and fungi in an ecosystem are important as
- A. decomposers that break down dead plants and animals
 - B. producers that make food from sunlight
 - C. abiotic components of the ecosystem
 - D. animals that feed on living plants
- 20.** If all the decomposers in a forest were removed, the MOST likely result would be that
- A. more rain would fall in the forest
 - B. all the animals would become plants
 - C. the forest would grow faster
 - D. dead plants and animals would pile up and nutrients would not return to the soil

- 21.** The non-living parts of an ecosystem are called
- A. biotic components
 - B. decomposers
 - C. abiotic components
 - D. consumers
- 22.** Which of the following is an abiotic component of a forest ecosystem?
- A. Earthworm
 - B. Climbing plant
 - C. Sunlight
 - D. Mushroom
- 23.** Which group contains ONLY abiotic components?
- A. Water, fish, soil, air
 - B. Water, air, soil, temperature
 - C. Air, water, bacteria, rock
 - D. Soil, grass, light, air
- 24.** Sunlight is an important abiotic component of an ecosystem chiefly because
- A. it keeps animals awake
 - B. it makes water salty
 - C. it helps the wind to blow
 - D. green plants need it to make food
- 25.** Water is important in an ecosystem because
- A. it prevents plants from growing too tall
 - B. it supplies the soil with sunlight
 - C. it is a living part of the ecosystem
 - D. all living things need it for their life processes
- 26.** A pond dries up completely during a long dry season. The fish die mainly because an abiotic component
- A. needed by the fish has been removed
 - B. has turned into a biotic component
 - C. has eaten the fish
 - D. has multiplied too fast
- 27.** Fish are suited to life in water because they
- A. have lungs and legs
 - B. have roots that take in water
 - C. have gills for breathing in water and fins for swimming
 - D. have wings and feathers

- 28.** Which structure helps a monkey to live in the branches of trees?
- A. A hard shell for protection
 - B. Long limbs and a gripping tail for holding on to branches
 - C. Gills for taking in air
 - D. Fins for moving about
- 29.** Water lilies are suited to life in a pond because they have
- A. deep woody roots
 - B. leaves reduced to spines
 - C. broad floating leaves that stay on the surface
 - D. thick bark on the stem
- 30.** Grasses of the northern savanna survive the long dry season mainly because they
- A. take in food from the air
 - B. can swim to wetter places
 - C. grow best without any water
 - D. have underground parts that sprout again when the rains come
- 31.** Which of the following animals is found in a terrestrial ecosystem?
- A. Grasscutter
 - B. Crab in a rock pool
 - C. Tilapia
 - D. Shrimp
- 32.** An earthworm taken from the soil and placed in a bucket of water soon dies. This shows that
- A. an organism survives best in the ecosystem to which it is suited
 - B. earthworms are not living things
 - C. every animal can live anywhere
 - D. water is harmful to all animals
- 33.** The MAIN difference between the organisms of a pond and those of a grassland is that
- A. pond organisms are all larger
 - B. each group has features suited to the conditions of its own surroundings
 - C. grassland organisms do not need water
 - D. pond organisms do not need food
- 34.** The feeding relationship in which a hawk feeds on a chick is an example of
- A. predation
 - B. competition
 - C. decomposition
 - D. pollination

- 35.** Two mango trees growing very close together struggle for light, water and soil nutrients. This is an example of
- A. competition
 - B. predation
 - C. decomposition
 - D. pollination
- 36.** Which of the following shows a dependence of a plant on an animal?
- A. A plant taking in water through its roots
 - B. A plant using sunlight to make food
 - C. A plant losing water from its leaves
 - D. A bee carrying pollen from one flower to another
- 37.** Green plants in an ecosystem supply animals with
- A. wind and rain
 - B. soil and rock
 - C. food and oxygen
 - D. light and heat
- 38.** Which of the following best shows how a biotic component depends on an abiotic component?
- A. A tree takes up water from the soil
 - B. A bird builds a nest with twigs
 - C. A fungus grows on a dead log
 - D. A goat feeds on grass
- 39.** A factory pours waste into a river and most of the fish die. The farmers who catch the fish also suffer. This shows that
- A. fish are not part of the ecosystem
 - B. rivers can clean themselves at once
 - C. the parts of an ecosystem are linked, so a change in one affects the others
 - D. only the fish are affected by pollution
- 40.** Which of the following human activities would MOST help to protect a wetland ecosystem near a town?
- A. Catching all the fish before the rains
 - B. Stopping the dumping of refuse in the wetland and replanting the plants around it
 - C. Filling the wetland with sand for building
 - D. Burning the plants at the edge of the water

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 7: Electricity and E

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Electricity produced by burning fuels such as natural gas, crude oil or coal is called
 - A. wind electricity
 - B. nuclear electricity
 - C. solar electricity
 - D. thermal electricity
2. In a thermal power station, the heat from the burning fuel is used to
 - A. cool the generator
 - B. melt the metal of the turbine
 - C. boil water and produce steam
 - D. charge large dry cells directly
3. In both thermal and nuclear power stations, the part that is turned by steam is the
 - A. pylon
 - B. transformer
 - C. turbine
 - D. chimney
4. The machine that changes the movement of the turbine into electricity is the
 - A. boiler
 - B. resistor
 - C. switch
 - D. generator
5. Nuclear electricity is produced from the heat released when
 - A. water falls from a high dam
 - B. coal is burnt in a large furnace
 - C. the nuclei of atoms of a fuel such as uranium are split
 - D. the Sun shines on a panel

- 6.** Which of the following is the fuel used in a nuclear power station?
- A. Uranium
 - B. Crude oil
 - C. Natural gas
 - D. Charcoal
- 7.** The energy change that takes place in the generator of a power station is
- A. chemical energy to heat energy
 - B. kinetic energy to electrical energy
 - C. electrical energy to kinetic energy
 - D. light energy to electrical energy
- 8.** Which statement about thermal and nuclear power stations is correct?
- A. Both burn fuel in a furnace to obtain heat
 - B. Both use steam to turn a turbine, but they obtain their heat in different ways
 - C. Both of them depend on falling water
 - D. Neither of them uses a generator
- 9.** The gas given off in the largest amount when fossil fuels are burnt in a thermal power station is
- A. nitrogen
 - B. oxygen
 - C. carbon dioxide
 - D. helium
- 10.** Carbon dioxide released from thermal power stations is of concern because it
- A. adds to the greenhouse gases that cause global warming
 - B. turns rain water into fresh water
 - C. makes the soil more fertile
 - D. destroys the ozone layer at once
- 11.** Smoke and soot from a thermal power station affect people living nearby mainly by
- A. improving the quality of their drinking water
 - B. causing breathing problems such as coughs and asthma
 - C. increasing the amount of oxygen in the air
 - D. making their houses stronger
- 12.** Sulphur dioxide released when some fuels are burnt may cause
- A. acid rain
 - B. a water cycle
 - C. an earthquake
 - D. a rainbow

- 13.** The MAIN environmental danger of a nuclear power station is
- A. the smoke from its chimney
 - B. the radioactive waste it produces and the risk of leakage
 - C. the carbon dioxide it releases
 - D. the noise of its boiler
- 14.** Which of the following would reduce the harmful effects of electricity generation on the environment?
- A. Building more chimneys for the same station
 - B. Pouring hot water from the station into a nearby river
 - C. Burning more coal instead of natural gas
 - D. Using more solar and hydro-electric power and fitting filters to chimneys
- 15.** Hot water released from a power station into a river may harm the river because it
- A. makes the river flow more quickly
 - B. raises the temperature of the water and kills fish and other organisms
 - C. removes all the salt from the water
 - D. adds useful nutrients to the water
- 16.** The letters LED stand for
- A. light energy detector
 - B. light-emitting diode
 - C. long electric drive
 - D. low energy device
- 17.** The component that gives out light when current passes through it in the correct direction is the
- A. LED
 - B. resistor
 - C. inductor
 - D. capacitor
- 18.** The MAIN function of a resistor in an electronic circuit is to
- A. reduce (limit) the current flowing in the circuit
 - B. change the current into heat only
 - C. give out light
 - D. store electrical charge
- 19.** The value of a small resistor is shown on its body by
- A. its length
 - B. the number of legs it has
 - C. its weight
 - D. coloured bands

- 20.** The component that stores electrical charge and releases it later is the
- A. switch
 - B. resistor
 - C. capacitor
 - D. diode
- 21.** A coil of wire that opposes a change in the current flowing through it is the
- A. inductor
 - B. LED
 - C. capacitor
 - D. resistor
- 22.** Which component allows current to pass in one direction only?
- A. A resistor
 - B. A connecting wire
 - C. A diode
 - D. A capacitor
- 23.** A source of electrical energy is needed in every working circuit. In the circuit built in class, this is provided by the
- A. resistor
 - B. LED
 - C. switch
 - D. dry cells
- 24.** Two dry cells of 1.5 V each are joined in series. The voltage of the battery formed is
- A. 3 V
 - B. 0.75 V
 - C. 6 V
 - D. 1.5 V
- 25.** A P-N junction diode is made by joining
- A. two pieces of copper wire
 - B. a resistor and a capacitor
 - C. a P-type region and an N-type region of semiconductor material
 - D. two dry cells in series
- 26.** In a P-N junction diode, the P region is the
- A. region that stores charge
 - B. positive region, which must be joined to the positive side of the battery for current to flow
 - C. region that gives off light
 - D. negative region, which must be joined to the positive side of the battery

- 27.** A diode connected so that current flows through it is said to be
- A. short circuited
 - B. reverse biased
 - C. earthed
 - D. forward biased
- 28.** An LED in a circuit does not light when the cells are connected one way round, but lights when the cells are turned round. This is because the LED
- A. needs a larger resistor
 - B. stores charge before lighting
 - C. allows current to pass in one direction only
 - D. is partly spoilt
- 29.** The longer leg of a new LED is the
- A. leg that must be cut off before use
 - B. positive leg (anode), which is joined to the positive side of the battery
 - C. negative leg, which is joined to the positive side of the battery
 - D. leg that carries no current
- 30.** When the switch in a circuit is closed,
- A. the circuit is broken and no current flows
 - B. the LED stores charge
 - C. the circuit is complete and current flows
 - D. the cells are destroyed
- 31.** When the switch in a circuit is opened, the LED goes out because
- A. the cells have run down at once
 - B. the path of the current is broken
 - C. the resistor has melted
 - D. the LED has been spoilt by the current
- 32.** In a circuit diagram, a long thin line beside a short thick line is the symbol for
- A. a cell
 - B. a resistor
 - C. a switch
 - D. a lamp
- 33.** Before joining an LED to a 3 V battery, it is wise to join a resistor in the circuit in order to
- A. prevent too large a current from spoiling the LED
 - B. store charge for the LED
 - C. make the LED brighter than normal
 - D. change the direction of the current

- 34.** A pupil built a circuit with two cells, a switch, a resistor and an LED, but the LED did not light even when the switch was closed. The FIRST thing she should check is
- A. whether the LED is warm
 - B. whether the wires are long enough
 - C. whether the room is dark enough
 - D. whether all the connections are tight and the cells are the right way round
- 35.** In a simple series circuit, the components are joined
- A. with the LED outside the circuit
 - B. in no particular order, since current finds its own way
 - C. side by side, so that each carries a different current
 - D. one after another, so that the same current passes through each of them
- 36.** When a second resistor is added in series to a circuit containing an LED, the LED becomes
- A. brighter, because the resistance is reduced
 - B. dimmer, because the voltage of the cells increases
 - C. dimmer, because the current is reduced
 - D. brighter, because the current is increased
- 37.** If the resistor in an LED circuit is replaced by one of much smaller resistance, the LED will
- A. go out completely
 - B. store more charge
 - C. change the direction of the current
 - D. shine more brightly and may be damaged by the larger current
- 38.** A diode joined into an LED circuit the wrong way round will make the LED
- A. shine more brightly
 - B. flash on and off
 - C. shine with a different colour
 - D. stay off, because the diode blocks the current
- 39.** The components of an electronic circuit are said to be interdependent because
- A. the way one component behaves affects the working of the others
 - B. each component works on its own
 - C. they are all made of the same material
 - D. they are all the same size

- 40.** A radio set, a mobile phone and a television all contain resistors, diodes and capacitors. This shows that
- A. all these gadgets do the same work
 - B. the same basic components are used in many different electronic gadgets
 - C. electronic gadgets need no source of energy
 - D. the components work only in radios

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a circuit diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 8: Energy

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Energy is best described as
 - A. the speed of a moving body
 - B. the space a body occupies
 - C. the mass of a body
 - D. the ability to do work
2. The unit in which energy is measured is the
 - A. newton
 - B. joule
 - C. kilogram
 - D. watt
3. The energy possessed by a body because of its position above the ground is
 - A. sound energy
 - B. potential energy
 - C. kinetic energy
 - D. nuclear energy
4. The energy possessed by a body because of its motion is
 - A. light energy
 - B. potential energy
 - C. chemical energy
 - D. kinetic energy
5. The energy stored in food, fuel and a dry cell is
 - A. solar energy
 - B. kinetic energy
 - C. sound energy
 - D. chemical energy
6. The energy that reaches the Earth directly from the Sun is called
 - A. solar energy
 - B. chemical energy
 - C. potential energy
 - D. nuclear energy

- 7.** Which form of energy reaches our ears when a drum is beaten?
- A. Light energy
 - B. Sound energy
 - C. Chemical energy
 - D. Nuclear energy
- 8.** Which of the following is NOT a form of energy?
- A. Heat
 - B. Light
 - C. Sound
 - D. Mass
- 9.** The sum of the potential energy and the kinetic energy of a body is called its
- A. nuclear energy
 - B. electrical energy
 - C. chemical energy
 - D. mechanical energy
- 10.** A coconut hanging on a tree possesses
- A. potential energy
 - B. sound energy
 - C. no energy at all
 - D. kinetic energy only
- 11.** As a mango falls from a tree to the ground, its
- A. potential energy increases while its kinetic energy decreases
 - B. potential and kinetic energy both increase
 - C. potential energy decreases while its kinetic energy increases
 - D. potential and kinetic energy both decrease
- 12.** A stone thrown straight upwards has its greatest potential energy
- A. half-way up
 - B. just before it hits the ground
 - C. at the highest point of its flight
 - D. at the moment it leaves the hand
- 13.** At the highest point of its flight, a ball thrown straight upwards has
- A. maximum potential energy and least kinetic energy
 - B. equal potential and kinetic energy
 - C. neither potential nor kinetic energy
 - D. maximum kinetic energy and least potential energy

- 14.** A pendulum bob swings from side to side. At the lowest point of the swing the bob has
- A. more potential than kinetic energy
 - B. no energy at all
 - C. the greatest kinetic energy
 - D. the greatest potential energy
- 15.** When a mango falls from a tree, the potential energy it loses is changed mainly into
- A. kinetic energy
 - B. nuclear energy
 - C. electrical energy
 - D. chemical energy
- 16.** A torch (flashlight) changes
- A. potential energy into sound energy
 - B. light energy into chemical energy
 - C. chemical energy into light energy
 - D. sound energy into heat energy
- 17.** An electric iron changes electrical energy into
- A. heat energy
 - B. chemical energy
 - C. nuclear energy
 - D. sound energy
- 18.** A solar panel on the roof of a clinic changes
- A. solar energy into electrical energy
 - B. heat energy into sound energy
 - C. electrical energy into solar energy
 - D. chemical energy into light energy
- 19.** A radio set changes electrical energy mainly into
- A. potential energy
 - B. chemical energy
 - C. sound energy
 - D. nuclear energy
- 20.** Which appliance changes electrical energy into kinetic energy?
- A. An electric fan
 - B. A candle
 - C. An electric bulb
 - D. A dry cell

- 21.** Water stored in the Akosombo dam high above the turbines possesses
- A. chemical energy, which is changed into heat energy
 - B. potential energy, which is changed into electrical energy at the power station
 - C. sound energy, which is changed into light energy
 - D. nuclear energy, which is changed into kinetic energy
- 22.** A burning candle changes chemical energy into
- A. sound and nuclear energy
 - B. heat and light energy
 - C. potential and sound energy
 - D. electrical and chemical energy
- 23.** The potential energy of a body is calculated using the expression
- A. $PE = mgh$
 - B. $PE = mv$
 - C. $PE = mg/h$
 - D. $PE = 1/2 mv^2$
- 24.** The kinetic energy of a body is calculated using the expression
- A. $KE = 2mv$
 - B. $KE = mgh$
 - C. $KE = mv$
 - D. $KE = 1/2 mv^2$
- 25.** A bag of mass 2 kg is lifted onto a shelf 5 m above the ground. Its potential energy is ($g = 10 \text{ m/s}^2$)
- A. 25 J
 - B. 1 000 J
 - C. 100 J
 - D. 10 J
- 26.** A boy of mass 40 kg runs at a speed of 3 m/s. His kinetic energy is
- A. 120 J
 - B. 180 J
 - C. 360 J
 - D. 60 J
- 27.** Two stones are held at the same height. The stone with the greater mass has
- A. no potential energy
 - B. the smaller potential energy
 - C. the same potential energy as the lighter stone
 - D. the greater potential energy

- 28.** If the speed of a moving car is doubled while its mass stays the same, its kinetic energy becomes
- A. half as large
 - B. the same
 - C. twice as large
 - D. four times as large
- 29.** The transfer of heat through a solid from the hot end to the cold end is called
- A. evaporation
 - B. conduction
 - C. radiation
 - D. convection
- 30.** Heat is transferred through liquids and gases mainly by
- A. radiation
 - B. convection
 - C. reflection
 - D. conduction
- 31.** The transfer of heat from the Sun to the Earth takes place by
- A. condensation
 - B. conduction
 - C. radiation
 - D. convection
- 32.** Which of the following is the BEST conductor of heat?
- A. Rubber
 - B. Aluminium
 - C. Plastic
 - D. Wood
- 33.** The handles of cooking pots are often made of wood or plastic because these materials are
- A. heavier than metal
 - B. good conductors of heat
 - C. poor conductors of heat
 - D. able to store heat for long
- 34.** Warm air above a hot road rises. This is because warm air
- A. has no weight at all
 - B. is pushed up by sound waves
 - C. is lighter (less dense) than the cooler air around it
 - D. is heavier than the cooler air

- 35.** The formation of a sharp shadow behind an object placed in sunlight shows that light
- A. is made up of sound waves
 - B. travels in a straight line
 - C. travels only in water
 - D. bends round objects
- 36.** Three cardboard sheets, each with a small hole, are arranged so that a candle flame can be seen through them. The flame can be seen only when the three holes are
- A. covered with glass
 - B. placed at different heights
 - C. of different sizes
 - D. in a straight line
- 37.** The bouncing back of light when it strikes a smooth, shiny surface is called
- A. refraction
 - B. dispersion
 - C. absorption
 - D. reflection
- 38.** A pencil standing in a glass of water appears bent at the surface of the water. This is due to
- A. refraction of light
 - B. absorption of light
 - C. reflection of light
 - D. dispersion of light
- 39.** The splitting of white light into its colours as it passes through a glass prism is called
- A. dispersion
 - B. conduction
 - C. reflection
 - D. diffusion
- 40.** A rainbow seen in the sky after a shower of rain is formed because the raindrops
- A. absorb all the colours of sunlight
 - B. reflect sound back to the Earth
 - C. split the sunlight into its different colours
 - D. give out light of their own

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in every calculation. Take the acceleration due to gravity, g , as 10 m/s^2 .

Question 7

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 9: Farming Systems

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A farming system is best described as
 - A. the way a farmer arranges and manages the growing of crops and the keeping of animals
 - B. the tools used to clear a piece of land
 - C. the number of workers on a farm
 - D. the type of soil found on a farm
2. Which of the following is NOT one of the farming systems studied at this level?
 - A. Cold storage farming
 - B. Organic farming
 - C. Crop rotation
 - D. Mixed farming
3. A farmer chooses a farming system mainly according to
 - A. the day of the week on which he starts
 - B. the number of children in his house
 - C. the colour of his farm tools
 - D. the land, the crops, the animals and the money available to him
4. Growing crops and keeping animals on the same farm at the same time is called
 - A. land rotation
 - B. mixed farming
 - C. mixed cropping
 - D. crop rotation
5. Growing two or more crops together on the same piece of land at the same time is called
 - A. mixed cropping
 - B. mixed farming
 - C. crop rotation
 - D. monocropping

- 6.** Growing only one type of crop on a piece of land season after season is called
- A. mixed farming
 - B. organic farming
 - C. mixed cropping
 - D. monocropping (sole cropping)
- 7.** In land rotation, the farmer
- A. uses only manure and no chemicals
 - B. keeps animals and crops on the same land
 - C. grows different crops on the same land each year
 - D. cultivates one piece of land for a few years, then leaves it and moves to another piece
- 8.** Land rotation is also known as
- A. crop rotation
 - B. shifting cultivation
 - C. plantation farming
 - D. mixed farming
- 9.** Land that has been left uncultivated so that it can regain its fertility is said to be under
- A. rotation of crops
 - B. fallow
 - C. mulch
 - D. irrigation
- 10.** During the fallow period the fertility of the land is restored mainly by
- A. the fertilizer spread before the land is left
 - B. the heat of the Sun on the bare soil
 - C. the ploughing done by the farmer
 - D. the natural vegetation that grows and decays on it
- 11.** Land rotation is practised mostly in areas where
- A. land is scarce and the population is large
 - B. the soil is already very rich
 - C. there are many tractors and fertilizers
 - D. land is plentiful and the population is small
- 12.** Which of the following is a disadvantage of land rotation?
- A. It gives the land no time to rest
 - B. It needs a large area of land and the burning of bush destroys soil organisms
 - C. It cannot be practised without a tractor
 - D. It makes the farmer grow only one crop

- 13.** Land rotation is becoming less common in many parts of Ghana today mainly because
- A. the government has banned all farming
 - B. the soil has become too rich for it
 - C. the growing population has reduced the amount of land available
 - D. farmers no longer know how to practise it
- 14.** In crop rotation, the farmer
- A. grows different crops on the same piece of land in a planned order, season after season
 - B. keeps animals instead of growing crops
 - C. moves to a new piece of land each year
 - D. grows the same crop on the land every year
- 15.** The MAIN reason for including a legume such as groundnut in a crop rotation is that the legume
- A. takes all the pests away
 - B. adds nitrogen to the soil
 - C. dries the soil quickly
 - D. gives the largest harvest
- 16.** The nitrogen added to the soil by a groundnut crop comes from
- A. the leaves falling on the soil
 - B. bacteria living in nodules on its roots
 - C. the water the plant takes in
 - D. the sunlight trapped by the plant
- 17.** Crop rotation helps to control pests and diseases because
- A. the pests of one crop do not find their food plant in the following season
 - B. the crops are sprayed with more chemicals
 - C. the pests are frightened by the new crops
 - D. the land is burnt before every planting
- 18.** In a good crop rotation plan, a deep-rooted crop is followed by a shallow-rooted one so that
- A. the soil becomes harder
 - B. only one type of nutrient is used up
 - C. the plants draw nutrients from different depths of the soil
 - D. the farmer can harvest at the same time
- 19.** Which of the following is the MAIN difference between crop rotation and land rotation?
- A. In land rotation the crops are changed each season
 - B. In crop rotation the same land is used and the crops are changed; in land rotation the land itself is changed

- C. In crop rotation only one crop is grown
 - D. Crop rotation can be practised only with animals
- 20.** A farmer at Wenchi grows maize, then beans, then cassava and then rests the land on the same plot year after year. This farmer is practising
- A. crop rotation
 - B. monocropping
 - C. mixed cropping
 - D. land rotation
- 21.** Which of the following is an example of mixed cropping?
- A. Maize, cassava and cowpea growing together in the same field
 - B. Maize this year and cassava on the same field next year
 - C. Maize grown alone on a large field
 - D. Maize grown on one field and goats kept on another
- 22.** One advantage of mixed cropping is that
- A. harvesting is always done by machine
 - B. the crops never compete for anything
 - C. the farmer still has something to harvest if one crop fails
 - D. only one type of nutrient is removed from the soil
- 23.** Mixed cropping helps to protect the soil because
- A. the different crops cover the soil and reduce erosion
 - B. the soil is left bare between the plants
 - C. the farmer ploughs the soil more often
 - D. the crops are burnt after harvesting
- 24.** A disadvantage of mixed cropping is that
- A. the soil cannot hold water at all
 - B. no crop can be grown with another
 - C. the crops may compete with one another for light, water and nutrients
 - D. the farmer can harvest only once in ten years
- 25.** Mixed cropping is difficult to carry out with machines mainly because
- A. the crops are planted too deep
 - B. the farmer plants only in the dry season
 - C. machines cannot move on a farm
 - D. the crops are of different kinds and ripen at different times
- 26.** A farmer plants cowpea between his rows of maize. Apart from extra food, the cowpea also helps by
- A. covering the ground and adding nitrogen to the soil
 - B. making the maize grow taller than usual
 - C. preventing rain from reaching the soil
 - D. driving away all the birds

- 27.** Which of the following describes a farm practising mixed farming?
- A. A farm where only cattle are kept
 - B. A farm where the land is left to rest for five years
 - C. A farm at Ejura where maize is grown and cattle are also kept
 - D. A farm where only maize and cassava are grown together
- 28.** In mixed farming, the animals are useful to the crops mainly because they
- A. shade the young crops from the Sun
 - B. supply manure that improves the fertility of the soil
 - C. carry the seeds to the field
 - D. eat the pests of the crops at night
- 29.** In mixed farming, the crops are useful to the animals mainly because they
- A. provide medicine for the animals
 - B. supply water to the animals
 - C. keep the animals warm at night
 - D. supply feed such as crop residues and grain
- 30.** One financial advantage of mixed farming is that the farmer
- A. is paid by the government each month
 - B. earns income from both crops and animals during the year
 - C. does not need to buy any tools
 - D. pays no money for labour
- 31.** A disadvantage of mixed farming is that it
- A. gives the farmer only one source of income
 - B. needs more capital, more labour and more skill than a single enterprise
 - C. cannot be practised in Ghana
 - D. always destroys the soil
- 32.** Mixed farming is said to make good use of a farm's resources because
- A. the farmer buys everything from the market
 - B. nothing is ever harvested from the farm
 - C. the waste of one enterprise is used by the other
 - D. the animals are kept far away from the crops
- 33.** Organic farming is a system in which the farmer
- A. keeps animals in cages all year round
 - B. grows only one crop on a large farm
 - C. grows crops and keeps animals without using artificial fertilizers and chemical pesticides
 - D. uses only artificial fertilizers

- 34.** Which of the following would an organic farmer use to feed his crops?
- A. NPK fertilizer
 - B. Sulphate of ammonia
 - C. Weedicide
 - D. Compost and animal manure
- 35.** An organic farmer controls insect pests mainly by
- A. leaving the pests to multiply freely
 - B. burning the whole farm every month
 - C. using natural methods such as neem extract, hand picking and natural enemies
 - D. spraying strong chemical insecticides
- 36.** One advantage of organic farming is that
- A. it works only on very poor soils
 - B. it needs very little labour
 - C. the yields are always the highest possible
 - D. the food produced is free from chemical residues and the soil organisms are protected
- 37.** A disadvantage of organic farming is that
- A. yields are often lower and more labour is needed
 - B. the food produced cannot be eaten
 - C. it cannot be practised without machines
 - D. it destroys the structure of the soil
- 38.** Organic produce often sells at a higher price than other produce because
- A. the government fixes a higher price for it
 - B. it can be stored for ten years
 - C. many buyers are willing to pay more for food grown without chemicals
 - D. it is always larger in size
- 39.** A farmer with a small piece of land near Kumasi, little money and a family to feed all year would be BEST advised to practise
- A. leaving the land to fallow for five years
 - B. land rotation on several large plots
 - C. mixed cropping with manure from his own animals
 - D. monocropping of one crop on the whole land
- 40.** Which statement about the farming systems is correct?
- A. Organic farming cannot be combined with crop rotation
 - B. Only one system may be used on a farm
 - C. Mixed farming cannot include any crops
 - D. More than one system can be used on the same farm at the same time

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 10: Force and Motion

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A force is best described as
 - A. the speed of a moving body
 - B. the mass of a body at rest
 - C. a push or a pull acting on a body
 - D. the energy stored in a body
2. The unit in which force is measured is the
 - A. newton
 - B. kilogram
 - C. watt
 - D. joule
3. Two forces acting on a body are said to be balanced when they are
 - A. applied at different times
 - B. equal in size and act in the same direction
 - C. unequal and act in opposite directions
 - D. equal in size and act in opposite directions
4. When the forces acting on a body at rest are balanced, the body will
 - A. move with increasing speed
 - B. begin to move slowly
 - C. turn round and round
 - D. remain at rest
5. A body begins to move only when the forces acting on it are
 - A. unbalanced
 - B. very small
 - C. equal and opposite
 - D. balanced
6. Two pupils pull a box in opposite directions with forces of 30 N and 50 N. The box will move
 - A. in neither direction
 - B. in the direction of the 50 N force
 - C. at right angles to both forces
 - D. in the direction of the 30 N force

- 7.** The tendency of a body to resist a change in its state of rest or of motion is called
- A. gravity
 - B. friction
 - C. inertia
 - D. pressure
- 8.** Of the following bodies, the one with the greatest inertia is a
- A. bicycle
 - B. football
 - C. wheelbarrow
 - D. loaded lorry
- 9.** Newton's first law of motion states that a body continues in its state of rest or of uniform motion in a straight line unless
- A. an unbalanced force acts on it
 - B. it is made of metal
 - C. it is placed on a smooth surface
 - D. its mass is increased
- 10.** Newton's first law of motion is also known as the law of
- A. conservation of energy
 - B. inertia
 - C. gravity
 - D. friction
- 11.** A metal ball rolling on a smooth floor finally stops. The force that stops it is
- A. the push that started it
 - B. the inertia of the ball
 - C. the weight of the ball
 - D. friction between the ball and the floor
- 12.** According to Newton's first law, a ball rolling on a perfectly smooth surface with no opposing force would
- A. move in a circle
 - B. keep moving in a straight line for ever
 - C. rise into the air
 - D. stop after a short distance
- 13.** A body moving with uniform velocity has
- A. a force acting on it in the direction of motion only
 - B. a large unbalanced force acting on it
 - C. no unbalanced force acting on it
 - D. no mass

- 14.** Which of the following is the BEST example of inertia of rest?
- A. A ball that bounces off a wall
 - B. A lorry that is difficult to stop when moving
 - C. A heavy stone that is difficult to start moving
 - D. A magnet that attracts a pin
- 15.** The quantity that measures the inertia of a body is its
- A. weight
 - B. mass
 - C. volume
 - D. speed
- 16.** Passengers in a trolley jerk forward when the driver brakes suddenly because their bodies
- A. tend to continue moving forward
 - B. lose their inertia
 - C. are pushed forward by the brakes
 - D. become heavier at that moment
- 17.** Seat belts are worn in vehicles mainly to
- A. hold the passenger back when the vehicle stops suddenly
 - B. make the seat comfortable
 - C. reduce the speed of the vehicle
 - D. keep the passenger warm
- 18.** When a vehicle at rest starts off suddenly, the passengers are thrown backwards because the
- A. seats push the passengers backwards
 - B. engine pulls the passengers backwards
 - C. weight of the passengers increases
 - D. lower part of the body moves with the vehicle while the upper part stays at rest
- 19.** Knocking the handle of a hoe against a hard surface tightens the blade. This is because the
- A. handle pushes the blade upwards
 - B. blade continues to move downwards when the handle stops
 - C. blade becomes heavier on hitting the surface
 - D. blade loses its inertia
- 20.** A tablecloth can be pulled quickly from under a glass without the glass falling because the glass
- A. tends to stay at rest
 - B. has no mass
 - C. is firmly stuck to the cloth
 - D. is pushed upwards by the cloth

- 21.** Dust is removed from a mat by beating it because, when the mat is struck, the
- A. mat attracts the dust strongly
 - B. dust is burnt by the stick
 - C. dust gains weight and falls
 - D. mat moves away while the dust particles stay at rest and are left behind
- 22.** A magnet is a material that can
- A. attract all materials
 - B. attract iron and certain other materials
 - C. produce electricity by itself
 - D. attract all metals
- 23.** Which of the following materials will be attracted by a magnet?
- A. A copper wire
 - B. A wooden pencil
 - C. A plastic ruler
 - D. An iron nail
- 24.** A magnet that keeps its magnetism for a long time is described as a
- A. soft magnet
 - B. temporary magnet
 - C. permanent magnet
 - D. weak magnet
- 25.** The poles of a magnet are the parts where the magnetic force is
- A. weakest
 - B. completely absent
 - C. strongest
 - D. the same as at the middle
- 26.** When the north pole of one magnet is brought near the north pole of another magnet, the two magnets
- A. repel each other
 - B. attract each other
 - C. both lose their magnetism at once
 - D. stick together firmly
- 27.** The surest test that a bar of metal is a magnet is that it
- A. is made of steel
 - B. repels one end of a known magnet
 - C. attracts an iron nail
 - D. is shiny and hard

- 28.** A bar magnet hung freely by a thread finally comes to rest with its ends pointing
- A. in no fixed direction
 - B. east and west
 - C. north and south
 - D. upwards and downwards
- 29.** Which of the following is a use of a magnet in everyday life?
- A. Separating iron pieces from other waste at a scrap yard
 - B. Cutting metal sheets
 - C. Boiling water in a kettle
 - D. Measuring the mass of a body
- 30.** A simple machine is a device that
- A. creates energy for the user
 - B. makes work easier to do
 - C. works without any force
 - D. destroys the force applied to it
- 31.** Which of the following is NOT a simple machine?
- A. A pulley
 - B. An inclined plane
 - C. A wheel and axle
 - D. A dry cell
- 32.** The fixed point about which a lever turns is called the
- A. effort
 - B. load
 - C. fulcrum
 - D. pivot arm
- 33.** A lever in which the fulcrum lies between the effort and the load is a
- A. second-class lever
 - B. third-class lever
 - C. fourth-class lever
 - D. first-class lever
- 34.** A wheelbarrow is an example of a
- A. second-class lever
 - B. first-class lever
 - C. pulley
 - D. third-class lever

- 35.** In a pair of sugar tongs, the effort is applied between the fulcrum and the load. The tongs are therefore a
- A. second-class lever
 - B. wheel and axle
 - C. third-class lever
 - D. first-class lever
- 36.** The work done by the effort on a machine is called the
- A. work input
 - B. work output
 - C. efficiency
 - D. mechanical advantage
- 37.** The efficiency of a machine is the ratio of
- A. work input to work output, expressed as a percentage
 - B. load to effort, expressed in newtons
 - C. effort to distance, expressed in joules
 - D. work output to work input, expressed as a percentage
- 38.** A machine is supplied with 200 J of energy and does 150 J of useful work. Its efficiency is
- A. 133%
 - B. 75%
 - C. 35%
 - D. 50%
- 39.** No simple machine has an efficiency of 100% because some of the work input is used in
- A. increasing the mass of the machine
 - B. overcoming friction between the moving parts
 - C. creating new energy in the machine
 - D. destroying the load
- 40.** The efficiency of a machine can be improved by
- A. adding more load to the machine
 - B. removing the fulcrum
 - C. oiling the moving parts to reduce friction
 - D. making the parts rougher

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps in any calculation, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 11: Human Body

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Any substance taken into the body to provide energy, growth and health is called
 - A. waste
 - B. enzyme
 - C. food
 - D. mucus
2. Which of the following is NOT a food nutrient?
 - A. Vitamin
 - B. Sand
 - C. Carbohydrate
 - D. Protein
3. The nutrient that supplies the body with most of its energy is
 - A. protein
 - B. mineral salt
 - C. carbohydrate
 - D. vitamin
4. Foods such as beans, fish, meat and eggs are valuable in the diet mainly because they supply
 - A. energy only
 - B. protein for growth and repair of the body
 - C. roughage only
 - D. water only
5. A person who does not eat enough food over a long time will
 - A. store more energy in the body
 - B. grow taller than others
 - C. become stronger and more active
 - D. lose weight, become weak and fall ill easily

- 6.** Roughage (dietary fibre) is important in the diet because it
- A. helps the movement of food through the alimentary canal and prevents constipation
 - B. builds strong bones and teeth
 - C. is digested into amino acids
 - D. supplies most of the body's energy
- 7.** Two pupils of the same age are compared. One is active, alert and of normal weight; the other is thin, tired and often ill. The BEST conclusion is that the second pupil
- A. takes too much water
 - B. does too little homework
 - C. may not be getting enough of the right food
 - D. was born to be weak
- 8.** The long tube through which food passes from the mouth to the anus is called
- A. the alimentary canal
 - B. the windpipe
 - C. the blood vessel
 - D. the spinal cord
- 9.** The correct order of the parts of the alimentary canal is
- A. mouth, oesophagus, stomach, small intestine, large intestine, anus
 - B. mouth, oesophagus, small intestine, stomach, large intestine, anus
 - C. mouth, windpipe, stomach, small intestine, large intestine, anus
 - D. mouth, stomach, oesophagus, large intestine, small intestine, anus
- 10.** The tube that carries food from the mouth to the stomach is the
- A. duodenum
 - B. trachea
 - C. rectum
 - D. oesophagus
- 11.** Digestion begins in the
- A. large intestine
 - B. small intestine
 - C. mouth
 - D. stomach
- 12.** The part of the alimentary canal where most digested food is absorbed is the
- A. stomach
 - B. small intestine
 - C. large intestine
 - D. oesophagus

- 13.** The main work of the large intestine is to
- A. produce saliva
 - B. absorb amino acids
 - C. digest protein
 - D. absorb water from the undigested material
- 14.** Undigested material is passed out of the body through the
- A. lungs
 - B. anus
 - C. kidney
 - D. skin
- 15.** Food is pushed along the alimentary canal by
- A. the air we breathe in
 - B. the force of gravity alone
 - C. the beating of the heart
 - D. the squeezing movement of the muscles of its wall
- 16.** The teeth help in digestion by
- A. cutting and grinding the food into smaller pieces
 - B. adding enzymes to the food
 - C. turning starch into sugar
 - D. absorbing nutrients from the food
- 17.** Saliva is produced in the
- A. small intestine
 - B. stomach
 - C. liver
 - D. mouth
- 18.** The substance in saliva that acts on boiled yam or bread is an enzyme that begins the digestion of
- A. protein
 - B. fat
 - C. starch
 - D. vitamin
- 19.** The breaking down of food into smaller pieces by the teeth and the tongue, without any chemical change, is called
- A. physical (mechanical) digestion
 - B. absorption
 - C. assimilation
 - D. chemical digestion

- 20.** Boiled rice chewed for a long time begins to taste sweet. This is because
- A. the enzyme in saliva has changed some of the starch into sugar
 - B. the rice has started to go bad
 - C. sugar has been added by the tongue
 - D. the teeth produce sugar
- 21.** The digestion of protein, as in a piece of meat, begins in the
- A. rectum
 - B. large intestine
 - C. mouth
 - D. stomach
- 22.** The acid produced in the stomach helps digestion by
- A. absorbing the digested food
 - B. making the food sweeter
 - C. killing germs in the food and helping the enzymes to work
 - D. changing fats into sugar
- 23.** Bile, which is produced by the liver and helps in the digestion of palm oil, acts by
- A. absorbing the oil into the blood
 - B. changing the oil into starch
 - C. breaking the oil into tiny droplets so that enzymes can act on it
 - D. turning the oil into protein
- 24.** An orange eaten at breakfast supplies vitamin C. In the alimentary canal this vitamin is
- A. digested into amino acids
 - B. absorbed without being digested
 - C. changed into starch
 - D. passed out unchanged in the faeces
- 25.** The end product of the digestion of starch is
- A. glucose
 - B. amino acids
 - C. fatty acids
 - D. glycerol only
- 26.** Proteins are digested into
- A. simple sugars
 - B. amino acids
 - C. fatty acids and glycerol
 - D. minerals

- 27.** Oils and fats are digested into
- A. glucose
 - B. amino acids
 - C. fatty acids and glycerol
 - D. vitamins
- 28.** Which of the following pairs correctly matches a food with the end product of its digestion?
- A. Palm oil - glucose
 - B. Boiled yam - fatty acids
 - C. Egg - amino acids
 - D. Bread - amino acids
- 29.** A meal of kenkey and fried fish will produce which end products of digestion?
- A. Glucose and amino acids
 - B. Fatty acids only
 - C. Glucose only
 - D. Amino acids only
- 30.** The end products of digestion are described as 'simple' because they are
- A. sweet in taste
 - B. small enough to pass through the wall of the small intestine into the blood
 - C. produced only in the mouth
 - D. always liquid in form
- 31.** Which of the following statements about digestion is correct?
- A. All food is absorbed without any change
 - B. Large food substances are broken down into smaller ones that the body can absorb
 - C. Small food substances are joined into larger ones
 - D. Food is burnt into ash inside the stomach
- 32.** The passing of digested food through the wall of the small intestine into the blood is called
- A. ingestion
 - B. egestion
 - C. digestion
 - D. absorption

- 33.** The inner wall of the small intestine is folded and covered with millions of tiny finger-like structures. This arrangement
- A. makes the intestine shorter
 - B. increases the surface for absorbing digested food
 - C. helps to chew the food
 - D. produces the teeth of the body
- 34.** Digested food is carried from the small intestine to all parts of the body by
- A. the air passages
 - B. the blood
 - C. the nerves
 - D. the bones
- 35.** Glucose carried to the cells of the body is used mainly to
- A. build the walls of cells
 - B. make the hair grow dark
 - C. form the enamel of teeth
 - D. release energy during respiration
- 36.** A patient whose small intestine has been damaged will suffer mainly because
- A. food cannot be chewed properly
 - B. water cannot be drunk at all
 - C. digested food cannot be properly absorbed into the blood
 - D. saliva can no longer be produced
- 37.** Iodine solution added to boiled yam turns
- A. blue-black
 - B. purple
 - C. colourless
 - D. brick-red
- 38.** Benedict's solution added to a glucose solution and heated turns
- A. purple
 - B. milky white
 - C. blue-black
 - D. brick-red
- 39.** Sodium hydroxide followed by copper sulphate solution added to egg white gives a
- A. purple colour
 - B. blue-black colour
 - C. green colour
 - D. brick-red colour

- 40.** A piece of paper rubbed with a food leaves a mark through which light passes. The food contains
- A. protein
 - B. vitamin C
 - C. starch
 - D. oil or fat

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 12: Human Body

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The class of food that supplies the body with most of its energy is
 - A. carbohydrates
 - B. proteins
 - C. mineral salts
 - D. vitamins
2. Proteins are described as body-building foods because they are used for
 - A. cooling the blood
 - B. growth and the repair of worn-out tissues
 - C. supplying heat only
 - D. protecting the body against all diseases
3. Which of the following foods is the RICHEST source of protein?
 - A. Yam
 - B. Cassava
 - C. Rice
 - D. Beans
4. Fats and oils are important in the diet because they
 - A. supply energy and help to store it in the body
 - B. are digested in the mouth
 - C. supply water to the cells
 - D. cannot be stored in the body
5. Vitamins and mineral salts are needed in the diet mainly to
 - A. build the bones only
 - B. replace the blood lost daily
 - C. protect the body and keep it working properly
 - D. supply all the energy for work
6. Which nutrient is needed for the formation of strong bones and teeth?
 - A. Sodium chloride
 - B. Calcium
 - C. Starch
 - D. Iodine

- 7.** Iron is needed in the diet for the formation of
- A. bile in the liver
 - B. sweat in the skin
 - C. haemoglobin in red blood cells
 - D. enamel on the teeth
- 8.** A diet that contains all the classes of food in the correct proportions is described as a
- A. balanced diet
 - B. liquid diet
 - C. protein diet
 - D. staple diet
- 9.** A deficiency disease is a disease caused by
- A. the lack of a particular nutrient in the diet
 - B. eating too much of one food
 - C. a virus entering the body
 - D. drinking cold water
- 10.** Kwashiorkor in young children is caused by the lack of
- A. fibre in the diet
 - B. protein in the diet
 - C. carbohydrate in the diet
 - D. water in the diet
- 11.** Night blindness is caused by a lack of
- A. iron
 - B. vitamin C
 - C. vitamin D
 - D. vitamin A
- 12.** Scurvy, in which the gums bleed, is caused by a lack of
- A. vitamin K
 - B. vitamin C
 - C. vitamin A
 - D. calcium
- 13.** Rickets in children is caused by a lack of
- A. vitamin C and iron
 - B. iodine and sodium
 - C. protein and water
 - D. vitamin D and calcium

- 14.** Anaemia, in which a person is always weak and pale, is caused by a lack of
- A. vitamin C
 - B. fat
 - C. iron
 - D. iodine
- 15.** Goitre, a swelling in the neck, is caused by a lack of
- A. iodine
 - B. vitamin D
 - C. calcium
 - D. protein
- 16.** A child who eats only kenkey and sugar every day, with no fish, beans or eggs, is MOST likely to suffer from
- A. goitre
 - B. night blindness
 - C. scurvy
 - D. kwashiorkor
- 17.** A child with kwashiorkor usually shows
- A. a large swelling in front of the neck
 - B. bowed legs and a soft skull
 - C. a swollen belly and feet, with thin arms and reddish hair
 - D. bleeding gums and loose teeth
- 18.** Bowed legs and soft, weak bones in a young child are symptoms of
- A. scurvy
 - B. rickets
 - C. anaemia
 - D. night blindness
- 19.** A person who cannot see well in dim light is suffering from
- A. night blindness
 - B. goitre
 - C. rickets
 - D. scurvy
- 20.** Rickets is BEST prevented by eating foods rich in vitamin D and by
- A. bathing the child in warm water
 - B. giving the child plenty of sugar
 - C. allowing the child to play in the early morning sunshine
 - D. keeping the child indoors at all times

- 21.** The BEST way to prevent scurvy in a family is to
- A. boil all fruits before eating them
 - B. eat more fried yam
 - C. drink more sweetened drinks
 - D. include fresh fruits such as oranges and guava in the daily meals
- 22.** Which of these local foods would BEST help to prevent anaemia?
- A. Sugar-cane juice
 - B. Kontomire and liver
 - C. Plain white rice
 - D. Fried plantain only
- 23.** Deficiency diseases in a community are BEST prevented by
- A. educating families to eat balanced diets from locally available foods
 - B. giving injections to every child each week
 - C. boiling all drinking water twice
 - D. wearing nose masks at meal times
- 24.** A virus is BEST described as a
- A. worm that lives in the intestines
 - B. large single-celled animal found in water
 - C. type of bacterium that lives in the soil
 - D. very small organism that can multiply only inside the cells of a living host
- 25.** Which of the following diseases is caused by a virus?
- A. COVID-19
 - B. Ringworm
 - C. Cholera
 - D. Malaria
- 26.** Viral diseases cannot be cured by
- A. vaccination before infection
 - B. antibiotics
 - C. the body's own defence system
 - D. rest and good feeding
- 27.** COVID-19, Ebola and H1N1 are alike in that all three are
- A. spread only by mosquitoes
 - B. deficiency diseases of childhood
 - C. caused by viruses and spread from person to person
 - D. caused by bacteria in food

- 28.** COVID-19 is spread MAINLY through
- A. drinking water from a tap
 - B. droplets released when an infected person coughs, sneezes or talks
 - C. eating food that lacks protein
 - D. the bite of an infected mosquito
- 29.** Ebola is spread MAINLY through
- A. direct contact with the body fluids of an infected person
 - B. the bite of a housefly
 - C. breathing the air of a cold room
 - D. sharing a classroom desk
- 30.** A disease that spreads quickly from community to community and from country to country, affecting a very large number of people, is declared a
- A. deficiency disease
 - B. chronic disease
 - C. pandemic
 - D. hereditary disease
- 31.** Air travel helped COVID-19 to spread round the world quickly because
- A. aeroplanes produce the virus in their engines
 - B. the virus is carried in aviation fuel
 - C. the virus grows faster in an aeroplane
 - D. infected people moved from country to country before showing symptoms
- 32.** Common symptoms of COVID-19 include
- A. a swelling in front of the neck
 - B. bleeding gums and loose teeth
 - C. fever, dry cough, tiredness and loss of taste or smell
 - D. bowed legs and soft bones
- 33.** A person suspected of having a viral disease such as COVID-19 should FIRST
- A. visit many friends for advice
 - B. isolate himself and report to a health facility
 - C. buy antibiotics from a drug store
 - D. continue to attend school as usual
- 34.** Washing the hands with soap under running water helps to prevent COVID-19 because it
- A. removes and destroys virus particles picked up on the hands
 - B. makes the hands too cold for the virus
 - C. kills mosquitoes on the skin
 - D. adds vitamins to the skin

- 35.** Wearing a nose mask in a crowded place helps to control a viral disease because the mask
- A. kills the virus in the whole room
 - B. traps the droplets breathed out and keeps out those from other people
 - C. cools the air that is breathed in
 - D. supplies oxygen to the wearer
- 36.** Besides illness and death, a serious effect of a pandemic on a country is that
- A. the population of mosquitoes falls
 - B. the rainfall of the country increases
 - C. the soil of the country loses its fertility
 - D. schools and businesses close and the economy suffers
- 37.** The MOST important role of an individual in controlling a viral disease is to
- A. share medicines with neighbours
 - B. hide any illness from the health workers
 - C. observe the safety measures and report any symptoms early
 - D. wait for the government to act alone
- 38.** A useful role for community members during an outbreak is to
- A. drive infected persons out of the community
 - B. hold large gatherings to discuss the disease
 - C. stop all health workers from entering the area
 - D. educate one another and support those in isolation with food and care
- 39.** Which of the following is a role of government in managing a pandemic?
- A. Preventing health workers from giving information
 - B. Closing all health facilities to save money
 - C. Providing vaccines, testing centres and treatment facilities
 - D. Asking every family to make its own medicine
- 40.** A class wishes to educate the community about a viral disease. The MOST suitable material to design and produce is a
- A. list of the names of sick people in the town
 - B. chart of the prices of medicines
 - C. map of the roads in the district
 - D. poster showing the symptoms and the ways of preventing the disease

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 13: Life Cycle Ho

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The series of changes an organism passes through from the egg to the adult is called
 - A. a life cycle
 - B. a food chain
 - C. a life span
 - D. a water cycle
2. The change in form that an insect undergoes as it grows from the egg to the adult is known as
 - A. fertilisation
 - B. photosynthesis
 - C. metamorphosis
 - D. respiration
3. An insect whose life cycle has the four stages egg, larva, pupa and adult shows
 - A. incomplete metamorphosis
 - B. continuous growth
 - C. no metamorphosis
 - D. complete metamorphosis
4. The life cycle of an organism is described as a cycle because
 - A. the organism returns to the egg stage as it grows old
 - B. it takes exactly one year to finish
 - C. the adult produces eggs and the same stages begin again
 - D. the organism grows round in shape
5. Which of the following animals has a life cycle most like that of the housefly?
 - A. The mosquito
 - B. The goat
 - C. The lizard
 - D. The fowl

- 6.** How many stages are there in the life cycle of the housefly?
- A. Two
 - B. Three
 - C. Four
 - D. Five
- 7.** The correct order of the stages in the life cycle of the housefly is
- A. larva, egg, pupa, adult
 - B. egg, larva, pupa, adult
 - C. egg, larva, adult, pupa
 - D. egg, pupa, larva, adult
- 8.** The first stage in the life cycle of the housefly is the
- A. larva
 - B. adult
 - C. pupa
 - D. egg
- 9.** The young housefly that hatches from the egg is called
- A. a nymph
 - B. a larva
 - C. an adult
 - D. a pupa
- 10.** The larva of the housefly is commonly known as
- A. a caterpillar
 - B. a maggot
 - C. a tadpole
 - D. a grub-fly
- 11.** The stage of the housefly that does not feed and appears to be at rest is the
- A. larva
 - B. egg
 - C. pupa
 - D. adult
- 12.** The adult housefly comes out of the
- A. soil only
 - B. pupa
 - C. egg
 - D. larva

- 13.** The stage of the life cycle in which the housefly lays eggs is the
- A. larva stage
 - B. egg stage
 - C. pupa stage
 - D. adult stage
- 14.** A pupil arranged cut-outs of the stages of the housefly in this order: pupa, egg, adult, larva. The arrangement is wrong because
- A. the larva must come before the egg
 - B. the pupa and the adult are the same stage
 - C. the egg comes first and the pupa comes after the larva
 - D. the adult comes first in every life cycle
- 15.** The eggs of the housefly are usually laid
- A. on dry stones in the sun
 - B. inside clean drinking water
 - C. on the leaves of growing crops only
 - D. in rotting rubbish, manure and other decaying matter
- 16.** Which of the following best describes the larva of the housefly?
- A. A tiny white oval object laid in a group
 - B. A winged insect with six legs
 - C. A hard, dark, motionless case
 - D. A white, legless, worm-like creature
- 17.** During the pupa stage, the housefly
- A. feeds heavily on rubbish
 - B. changes inside a hard case into the adult form
 - C. flies about looking for food
 - D. lays many eggs
- 18.** Which stage of the housefly feeds most and grows fastest?
- A. The larva
 - B. The pupa
 - C. The egg
 - D. The adult
- 19.** The complete life cycle of the housefly may take only about one to three weeks. This is important because
- A. each housefly lives for many years
 - B. only a few eggs are laid at a time
 - C. the housefly cannot spread diseases quickly
 - D. very large numbers of houseflies can be produced in a short time

- 20.** The warm, moist conditions of a refuse dump affect the life cycle of the housefly by
- A. making the stages develop faster
 - B. making the adult unable to fly
 - C. stopping the pupa from forming
 - D. killing the eggs at once
- 21.** Houseflies are mostly found around
- A. refuse dumps, faeces and uncovered food
 - B. clean, dry and well-swept rooms
 - C. the open sea
 - D. deep wells of clear water
- 22.** Which of the following is NOT a feeding place of the housefly?
- A. An open pan of cooked rice
 - B. A dead animal by the roadside
 - C. A sealed tin of milk
 - D. A heap of animal manure
- 23.** The mouth parts of the housefly are suited for
- A. tearing flesh from bones
 - B. sucking up liquid and softened food
 - C. biting and chewing hard grains
 - D. piercing the skin and sucking blood
- 24.** A housefly feeding on solid sugar first
- A. swallows it whole
 - B. bites it into small pieces
 - C. warms it with its wings
 - D. pours saliva on it to dissolve it
- 25.** A housefly carries germs mainly on its
- A. eyes only
 - B. eggs only
 - C. legs, body hairs and mouth parts
 - D. wings only
- 26.** Houseflies move between a refuse dump and the food on a table. This habit is dangerous because
- A. the flies lay eggs that people can see
 - B. the flies make the food cold
 - C. the flies eat all the food
 - D. germs picked up from the refuse are left on the food

- 27.** A gutter behind a chop bar is blocked with food waste and water. The best reason why many houseflies are seen there is that the gutter
- A. has clean water for the flies to drink
 - B. provides food and a breeding place for the flies
 - C. keeps away the enemies of the flies
 - D. is warmer than the air around
- 28.** An insect that lives close to human beings and causes harm and discomfort is described as
- A. a predator
 - B. a menace
 - C. a decomposer
 - D. a pollinator
- 29.** Which of the following diseases can be spread by the housefly?
- A. Dysentery
 - B. Measles
 - C. Ringworm
 - D. Malaria
- 30.** Food that has been visited by houseflies may cause
- A. sickle cell disease
 - B. food poisoning
 - C. night blindness
 - D. asthma
- 31.** Apart from spreading disease, houseflies are a problem because they
- A. bite people at night and suck blood
 - B. destroy clothes in the wardrobe
 - C. are a nuisance in the home and environment
 - D. eat the roots of crops
- 32.** A child in a home where food is left uncovered keeps having diarrhoea. The most likely explanation is that
- A. houseflies have bitten the child
 - B. the food was cooked for too long
 - C. the child drinks too much boiled water
 - D. houseflies have carried germs from faeces or refuse onto the food
- 33.** Which of the following is the strongest evidence that houseflies spread disease in a community?
- A. Houseflies make a buzzing sound while flying
 - B. Houseflies fly faster in the afternoon
 - C. Houseflies are seen in every town
 - D. Communities with poor refuse disposal and many flies record more cases of diarrhoea

- 34.** Which disease of the ones listed is spread by the mosquito and NOT by the housefly?
- A. Malaria
 - B. Dysentery
 - C. Typhoid fever
 - D. Cholera
- 35.** The best single way to stop houseflies from breeding in a compound is to
- A. spray perfume in the rooms each morning
 - B. sweep the compound once a month
 - C. collect and dispose of refuse properly
 - D. keep the windows open all day
- 36.** Covering food with a net cover helps to control the housefly menace because it
- A. keeps the flies away from the food
 - B. prevents the eggs from hatching
 - C. stops the food from getting cold
 - D. kills the flies that come near
- 37.** Which of the following practices will NOT help to reduce the number of houseflies in a community?
- A. Throwing kitchen waste behind the house
 - B. Using a pit latrine with a tight cover
 - C. Clearing choked gutters regularly
 - D. Covering refuse containers with lids
- 38.** Faeces left in the open is a serious health problem mainly because
- A. it attracts rain to the area
 - B. it makes the soil too rich for crops
 - C. houseflies breed in it and carry the germs to food
 - D. it dries up quickly in the sun
- 39.** A Basic 7 class wants to teach their community how to reduce the housefly menace. The MOST effective plan would be to
- A. ask each pupil to kill ten flies at home
 - B. hold a clean-up of the refuse dump and gutters and give a talk on covering food and refuse
 - C. buy insecticide and spray the classroom only
 - D. write the information in their exercise books

- 40.** A household covers its food carefully but still throws its kitchen waste in an open heap beside the kitchen. The weakness of this arrangement is that
- A. the flies still breed in the heap and will reach the food whenever it is opened
 - B. covering food is of no use at all
 - C. the heap will stop the flies from laying eggs
 - D. the covered food will go bad faster

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 14: Living Cells

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The smallest unit of a living organism that can carry out all the activities of life is
 - A. the system
 - B. the cell
 - C. the organ
 - D. the tissue
2. The instrument used in the laboratory to see cells clearly is
 - A. the microscope
 - B. the barometer
 - C. the stethoscope
 - D. the thermometer
3. A thin piece of onion was placed on a glass slide and viewed with a microscope. The many small boxes seen were
 - A. drops of water
 - B. crystals of salt
 - C. grains of sugar
 - D. cells
4. Which of the following statements about living things is correct?
 - A. Only animals are made up of cells
 - B. Both plants and animals are made up of cells
 - C. Only plants are made up of cells
 - D. Neither plants nor animals are made up of cells
5. A group of similar cells working together to perform the same work forms
 - A. an organism
 - B. a tissue
 - C. an organelle
 - D. an organ

- 6.** The small structures found inside a cell, each doing a special work, are called
- A. systems
 - B. tissues
 - C. organs
 - D. organelles
- 7.** Which of the following parts is found in an animal cell?
- A. Cell membrane
 - B. Cell wall
 - C. Chloroplast
 - D. Large central vacuole
- 8.** The part of an animal cell that forms its outer boundary is the
- A. cell wall
 - B. nucleus
 - C. cell membrane
 - D. cytoplasm
- 9.** The jelly-like material in which the organelles of a cell float is the
- A. nucleus
 - B. cell sap
 - C. chloroplast
 - D. cytoplasm
- 10.** The largest organelle in a typical animal cell is usually the
- A. ribosome
 - B. mitochondrion
 - C. nucleus
 - D. vacuole
- 11.** Which of the following is NOT found in an animal cell?
- A. Mitochondrion
 - B. Cell wall
 - C. Cytoplasm
 - D. Nucleus
- 12.** The sausage-shaped organelles found in the cytoplasm of an animal cell are the
- A. nuclei
 - B. chloroplasts
 - C. mitochondria
 - D. cell walls

- 13.** An animal cell has an irregular shape MAINLY because
- A. it has no cell wall to hold it in a fixed shape
 - B. it has no cell membrane
 - C. it has no nucleus
 - D. its cytoplasm is too thick to move
- 14.** A cell was taken from the cheek lining of a goat. The cell would contain
- A. chloroplasts, a cell wall and a nucleus
 - B. a cell wall, cytoplasm and a large vacuole
 - C. a nucleus, cytoplasm and a cell membrane
 - D. a cell wall, chloroplasts and a large vacuole
- 15.** The part of the cell that controls all the activities of the cell is the
- A. nucleus
 - B. cytoplasm
 - C. ribosome
 - D. cell membrane
- 16.** The cell membrane is important because it
- A. stores the water of the cell
 - B. gives the cell its green colour
 - C. controls the substances that enter and leave the cell
 - D. makes food for the cell
- 17.** The organelle in which energy is released from food in a cell is the
- A. mitochondrion
 - B. cell membrane
 - C. vacuole
 - D. nucleus
- 18.** Most of the chemical reactions of a cell take place in the
- A. cell membrane
 - B. cytoplasm
 - C. nucleus
 - D. cell wall
- 19.** Proteins are built up in a cell at the
- A. chloroplasts
 - B. cell membrane
 - C. vacuoles
 - D. ribosomes

- 20.** A muscle cell has very many mitochondria. This is because the muscle cell
- A. has no nucleus of its own
 - B. needs to store a large amount of water
 - C. needs a large amount of energy
 - D. has to make its own food
- 21.** If the nucleus is removed from a living cell, the cell will MOST likely
- A. produce more energy than before
 - B. change into a plant cell
 - C. die, because its activities are no longer controlled
 - D. continue to grow faster than before
- 22.** The outermost part of a plant cell is the
- A. cell membrane
 - B. nucleus
 - C. cytoplasm
 - D. cell wall
- 23.** The cell wall of a plant cell is made MAINLY of
- A. cellulose
 - B. starch
 - C. protein
 - D. fat
- 24.** The green organelles found in the cells of a leaf are the
- A. mitochondria
 - B. ribosomes
 - C. nuclei
 - D. chloroplasts
- 25.** The green colouring matter found inside the chloroplast is
- A. haemoglobin
 - B. cell sap
 - C. cytoplasm
 - D. chlorophyll
- 26.** A large space filled with cell sap in the middle of a mature plant cell is the
- A. nucleus
 - B. cell wall
 - C. vacuole
 - D. chloroplast

- 27.** In a mature plant cell the nucleus is usually found
- A. outside the cell wall
 - B. exactly in the centre of the cell
 - C. inside the large vacuole
 - D. pushed to one side, near the cell membrane
- 28.** Which part of a plant cell would NOT be found in a cell taken from the root of the plant?
- A. Vacuole
 - B. Chloroplast
 - C. Nucleus
 - D. Cell wall
- 29.** The chief function of the cell wall is to
- A. release energy from food
 - B. control what enters and leaves the cell
 - C. give the cell a fixed shape and support
 - D. make food for the plant
- 30.** Food is made in the plant cell in the
- A. chloroplasts
 - B. nucleus
 - C. mitochondria
 - D. vacuole
- 31.** The large vacuole of a plant cell is important because it
- A. makes the proteins of the cell
 - B. stores cell sap and keeps the cell firm
 - C. controls the activities of the cell
 - D. releases energy from food
- 32.** A plant that has not been watered for several days becomes limp MAINLY because
- A. the vacuoles of its cells have lost water
 - B. the cell walls have been destroyed
 - C. the chloroplasts have lost their colour
 - D. the nuclei of the cells have died
- 33.** The cell membrane of a plant cell lies
- A. just inside the cell wall
 - B. just outside the cell wall
 - C. inside the large vacuole
 - D. inside the nucleus

- 34.** A plant cell placed in the dark for a long time would be unable to
- A. keep its shape
 - B. make its own food
 - C. release energy from stored food
 - D. take in water
- 35.** Which of the following parts is found in a plant cell but NOT in an animal cell?
- A. Nucleus
 - B. Cell membrane
 - C. Cytoplasm
 - D. Chloroplast
- 36.** Which parts are found in BOTH plant and animal cells?
- A. Chloroplast, vacuole and cell wall
 - B. Nucleus, cytoplasm and cell membrane
 - C. Cell wall, nucleus and chloroplast
 - D. Cell wall, cytoplasm and chloroplast
- 37.** A pupil looked at a cell under a microscope and saw a cell wall, chloroplasts and a large central vacuole. The cell was
- A. a red blood cell
 - B. a plant cell
 - C. a cheek cell
 - D. an animal cell
- 38.** Plant cells usually have a regular box-like shape while animal cells do not. The reason is that the plant cell has
- A. a firm cell wall outside the cell membrane
 - B. no cytoplasm in its cell
 - C. a larger nucleus than the animal cell
 - D. more mitochondria than the animal cell
- 39.** A pupil was asked to model a plant cell and an animal cell with local materials. The BEST way to show the difference between them would be to
- A. use a larger nucleus in the plant cell model
 - B. colour both models green
 - C. leave out the cytoplasm in the animal cell model
 - D. put a firm box round the model of the plant cell only
- 40.** Which observation would BEST convince a class that both plants and animals are made up of cells?
- A. Feeling the surface of a leaf and the skin of the arm
 - B. Looking at a chart of a plant cell only
 - C. Viewing slides of leaf tissue and cheek tissue and seeing similar small

units in both

- D. Reading in a textbook that living things are made of cells

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 15: Materials

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Anything that has mass and occupies space is called
 - A. a force
 - B. a shadow
 - C. matter
 - D. energy
2. The three states in which materials are commonly found are
 - A. wood, metal and glass
 - B. hard, soft and rough
 - C. natural, artificial and waste
 - D. solid, liquid and gas
3. A pupil described a material as hard, rough and grey, and said that it kept its own shape. The material is MOST likely
 - A. palm oil
 - B. a stone
 - C. cooking gas
 - D. kerosene
4. Which of the following properties would a pupil record by LOOKING at a material only?
 - A. Its smell
 - B. Its colour
 - C. Its temperature
 - D. Its mass
5. Which group contains only solids?
 - A. Chalk, sand, iron nail
 - B. Sand, petrol, oxygen
 - C. Chalk, water, iron nail
 - D. Iron nail, air, palm oil

- 6.** Which of the following is a liquid at room temperature?
- A. Oxygen
 - B. Charcoal
 - C. Groundnut oil
 - D. Common salt
- 7.** A material that has no colour, cannot be held in the hand and spreads to fill the whole room is
- A. a liquid
 - B. a gas
 - C. a solid
 - D. a mixture of solids
- 8.** A material that has a fixed shape and a fixed volume is
- A. a solid
 - B. a gas
 - C. a vapour
 - D. a liquid
- 9.** A liquid has
- A. both a fixed shape and a fixed volume
 - B. a fixed shape but no fixed volume
 - C. neither a fixed volume nor a fixed mass
 - D. a fixed volume but no fixed shape
- 10.** Water poured from a bucket into a bottle takes the shape of the bottle. This shows that a liquid
- A. has no mass of its own
 - B. has no shape of its own
 - C. has no volume of its own
 - D. changes into a gas in the bottle
- 11.** In which state of matter are the particles packed closely together in a regular pattern?
- A. Vapour
 - B. Gas
 - C. Liquid
 - D. Solid
- 12.** The particles of a gas are
- A. far apart and move about freely
 - B. joined in long chains that cannot move
 - C. close together but able to slide over one another
 - D. close together and fixed in position

- 13.** A cylinder of cooking gas feels heavy because
- A. a gas has no mass but the air presses on it
 - B. a gas has mass even though it cannot be seen
 - C. the gas has a fixed shape inside the cylinder
 - D. the gas has turned into a solid inside
- 14.** Which property is shared by solids, liquids and gases?
- A. They can all be compressed easily
 - B. They all have mass and occupy space
 - C. They all have a fixed shape
 - D. They can all be poured from one container into another
- 15.** The liquid that is most important for the survival of humans is
- A. kerosene
 - B. petrol
 - C. water
 - D. palm oil
- 16.** Which of the following is a use of water in the human body?
- A. It hardens the bones of the body
 - B. It changes food into oxygen
 - C. It helps in the digestion and transport of food
 - D. It makes the body produce light
- 17.** Blood is an important liquid in the body because it
- A. carries food and oxygen to all parts of the body
 - B. stores all the fat of the body
 - C. removes bones that are damaged
 - D. keeps the bones stiff and strong
- 18.** A farmer at Techiman uses a liquid obtained from the oil palm fruit for cooking. The liquid is
- A. palm oil
 - B. vinegar
 - C. kerosene
 - D. petrol
- 19.** Which of the following shows the need to preserve liquids in a community?
- A. Pouring left-over drinking water on the ground
 - B. Leaving the tap running while brushing the teeth
 - C. Covering the water in a storage drum and using it carefully
 - D. Washing the compound with pipe-borne water every morning

- 20.** A tanker of petrol is important to a town MAINLY because petrol is used
- A. for drinking during the dry season
 - B. as a fuel in vehicles and machines
 - C. for watering crops on a farm
 - D. for washing cooking utensils
- 21.** Which practice would BEST protect the liquid stored in a water tank at a school?
- A. Keeping the tank covered and cleaning it regularly
 - B. Siting the tank beside the refuse dump
 - C. Allowing pupils to dip their cups into the tank
 - D. Leaving the lid open so that air can enter
- 22.** Which solid material is used MAINLY for roofing buildings in many Ghanaian towns?
- A. Glass
 - B. Cotton wool
 - C. Rubber
 - D. Aluminium sheet
- 23.** A solid obtained from plants and used for making tables and chairs is
- A. clay
 - B. cement
 - C. iron
 - D. wood
- 24.** Common salt is an important solid in the home because it
- A. adds taste to food and supplies minerals the body needs
 - B. makes food cook without fire
 - C. keeps the kitchen cool
 - D. turns food into a liquid
- 25.** Which of the following solids is used by a mason to build a wall?
- A. Palm fronds
 - B. Rubber tyres
 - C. Cement blocks
 - D. Cotton cloth
- 26.** Iron is widely used for making farm tools such as cutlasses MAINLY because it is
- A. shiny and colourful
 - B. cheap and smooth
 - C. hard and strong
 - D. light and soft

- 27.** A Basic 7 class was asked to explain why useful solids in the environment should be preserved. The BEST reason is that
- A. they are found only in the forest zone
 - B. they are limited in supply and are needed by humans and other living things
 - C. they are too heavy to be carried away
 - D. they lose their colour when they are used
- 28.** A substance that cannot be broken down into simpler substances by a chemical method is
- A. an element
 - B. a mixture
 - C. an alloy
 - D. a solution
- 29.** The chemical symbol of sodium is
- A. So
 - B. Sn
 - C. S
 - D. Na
- 30.** The element whose chemical symbol is K is
- A. potassium
 - B. carbon
 - C. calcium
 - D. krypton
- 31.** Which of the following gives the correct symbol of calcium?
- A. Cl
 - B. C
 - C. Cm
 - D. Ca
- 32.** The first element in the periodic table is
- A. oxygen
 - B. helium
 - C. hydrogen
 - D. lithium
- 33.** The elements in the periodic table are arranged in order of increasing
- A. atomic number
 - B. the date on which the element was discovered
 - C. mass of one litre of the element
 - D. the first letter of the name of the element

- 34.** An element has atomic number 8. The element is
- A. sulphur
 - B. carbon
 - C. oxygen
 - D. neon
- 35.** Which of the following is a property of MOST metals?
- A. They break easily into powder
 - B. They are poor conductors of heat
 - C. They are always liquids at room temperature
 - D. They conduct heat and electricity
- 36.** Which of the following elements is a non-metal?
- A. Aluminium
 - B. Sulphur
 - C. Potassium
 - D. Magnesium
- 37.** Helium, neon and argon are grouped together in the periodic table because they
- A. burn easily in air
 - B. are all liquids at room temperature
 - C. are all metals
 - D. are unreactive gases
- 38.** Elements that are found in the same group of the periodic table
- A. have exactly the same mass
 - B. have similar chemical properties
 - C. are all metals
 - D. have the same atomic number
- 39.** Sodium, magnesium and aluminium are all found in Period 3 of the periodic table. This means that they
- A. belong to the same group
 - B. have the same number of protons
 - C. are in the same horizontal row of the table
 - D. are all noble gases
- 40.** A pupil found that a shiny element conducted electricity, could be beaten into sheets and did not break when it was bent. The element is BEST described as
- A. a mixture
 - B. a non-metal
 - C. a noble gas
 - D. a metal

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 16: Science and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A career in science is BEST described as
 - A. work that needs no training at all
 - B. work done only by teachers
 - C. any work done inside a building
 - D. work in which a person uses scientific knowledge and skills to earn a living
2. A scientist who studies living things such as plants and animals is a
 - A. meteorologist
 - B. geologist
 - C. biologist
 - D. physicist
3. The scientist who studies substances, how they are made up and how they react is a
 - A. biologist
 - B. astronomer
 - C. surveyor
 - D. chemist
4. A person who designs and supervises the building of roads, bridges and dams is a
 - A. veterinary officer
 - B. civil engineer
 - C. nurse
 - D. pharmacist
5. A scientist who treats sick animals on a farm is a
 - A. veterinary doctor
 - B. agronomist
 - C. pharmacist
 - D. radiographer

- 6.** The health worker who prepares and dispenses medicines in a hospital is the
- A. physiotherapist
 - B. pharmacist
 - C. midwife
 - D. laboratory technician
- 7.** A person who studies the weather and prepares the daily forecast is a
- A. food scientist
 - B. meteorologist
 - C. zoologist
 - D. geologist
- 8.** A scientist who works to improve the processing and preservation of foods such as cassava and maize is a
- A. civil engineer
 - B. astronomer
 - C. surveyor
 - D. food scientist
- 9.** Which of the following careers is MOST directly concerned with the study of rocks and minerals?
- A. Food science
 - B. Computer programming
 - C. Nursing
 - D. Geology
- 10.** Professor Francis Allotey, a Ghanaian scientist of international repute, worked chiefly in the field of
- A. surgery and nursing
 - B. music and drama
 - C. mathematics and physics
 - D. law and government
- 11.** Professor Ibok Oduro of KNUST is known for her work in
- A. food science and the use of local food crops
 - B. the mining of gold
 - C. the design of aeroplane engines
 - D. the printing of newspapers
- 12.** Professor Ewurama Addy, a Ghanaian woman scientist, is known for her work in
- A. forecasting the weather
 - B. repairing motor vehicles
 - C. building bridges
 - D. biochemistry and the biological sciences

- 13.** Professor Jophus Anamuah-Mensah is best known in Ghana as a
- A. science educator
 - B. military officer
 - C. professional footballer
 - D. textile trader
- 14.** Professors Theophilus Ossei-Anto and Christian Anthony-Krueger have both contributed to
- A. the building of the Akosombo Dam
 - B. the printing of currency notes
 - C. the teaching and learning of science in Ghana
 - D. the export of cocoa beans
- 15.** Studying the lives of Ghanaian scientists is useful to pupils chiefly because it
- A. helps them to pass examinations without studying
 - B. shows them that Ghanaians can succeed in science and gives them models to follow
 - C. shows that only professors can study science
 - D. proves that science began in Ghana
- 16.** A science educator is a person who
- A. sells science textbooks
 - B. repairs laboratory apparatus
 - C. studies and improves the way science is taught and learnt
 - D. builds laboratories for a living
- 17.** Which institution in Ghana is BEST known for training engineers and scientists?
- A. The Ghana Standards Authority library
 - B. The National Theatre
 - C. The Ghana Police Training School
 - D. Kwame Nkrumah University of Science and Technology
- 18.** Alexander Fleming is remembered for the discovery of
- A. the electric bulb
 - B. the law of gravity
 - C. penicillin, the first antibiotic
 - D. the structure of the atom
- 19.** Penicillin has improved human life chiefly because it
- A. cures bacterial infections that once killed many people
 - B. supplies vitamins to the body
 - C. prevents all accidents on the roads
 - D. removes waste from the environment

- 20.** Albert Einstein is famous for his work in
- A. the classification of plants
 - B. physics, especially the theory of relativity
 - C. the discovery of vaccines
 - D. the invention of the telephone
- 21.** Charles Darwin is remembered for his explanation of
- A. how electricity flows in a wire
 - B. how living things change over time by natural selection
 - C. how the heart pumps blood
 - D. how rain is formed
- 22.** Stephen Hawking was a scientist noted for his study of
- A. the design of motor cars
 - B. the growth of crops
 - C. the universe, including black holes
 - D. the treatment of malaria
- 23.** Louis Pasteur's work on germs led directly to
- A. the discovery of gold in rocks
 - B. the pasteurisation of milk and the use of vaccines
 - C. the building of the first computer
 - D. the invention of the aeroplane
- 24.** The MAIN lesson pupils can learn from the lives of great scientists is that
- A. a scientist must be born outside Africa
 - B. discoveries are made only by accident
 - C. science needs no experiments at all
 - D. careful observation, hard work and curiosity lead to discovery
- 25.** A pupil preparing a presentation on a named scientist should include
- A. the scientist's field of work, the discovery made and how it has helped people
 - B. only the scientist's date of birth
 - C. a list of the scientist's relatives
 - D. the scientist's favourite food
- 26.** Technology is BEST described as
- A. the study of living things only
 - B. the use of scientific knowledge to make useful things and solve problems
 - C. the collection of old machines
 - D. the writing of science examinations

- 27.** The use of mobile phones for money transfer in Ghana is an example of the impact of science and technology on
- A. the movement of the planets
 - B. the growth of trees in the forest
 - C. trade and daily life in the community
 - D. the rainfall of the country
- 28.** A refrigerator improves life in the home chiefly because it
- A. cooks food without fire
 - B. removes salt from food
 - C. makes food richer in protein
 - D. keeps food cold so that it does not spoil quickly
- 29.** The use of solar panels in schools without a reliable supply of electricity shows that science and technology can
- A. remove the need for teachers
 - B. make the Sun shine more brightly
 - C. provide practical solutions to local problems
 - D. increase the number of power cuts
- 30.** The use of tractors, improved seeds and irrigation on farms has MAINLY led to
- A. higher yields and more food for the population
 - B. a fall in the quality of all crops
 - C. the end of rainfall in farming areas
 - D. the loss of all farming jobs
- 31.** Which of the following shows the impact of science in the school?
- A. The sweeping of the compound each morning
 - B. The wearing of school uniform
 - C. The use of computers and projectors in teaching
 - D. The singing of the national anthem
- 32.** A NEGATIVE effect of some modern technology on the community is
- A. easier travel from town to town
 - B. pollution of the air and water by factories and vehicles
 - C. faster communication between people
 - D. better treatment of diseases
- 33.** Innovation in science means
- A. memorising scientific facts
 - B. copying a classmate's work
 - C. repeating exactly what others have done
 - D. putting a new or improved idea into practical use

- 34.** Ghana needs to train more agricultural scientists MAINLY in order to
- A. stop the export of cocoa
 - B. reduce the number of farms in the country
 - C. increase food production and reduce the importation of food
 - D. increase the price of fertiliser
- 35.** Training more medical and laboratory scientists in Ghana would MOST directly
- A. improve the diagnosis and treatment of diseases in the country
 - B. lower the quality of local foods
 - C. reduce the number of schools
 - D. increase the rainfall of the country
- 36.** A reason for Ghana to focus on careers in renewable energy is that they would help the country to
- A. use up its crude oil more quickly
 - B. produce cleaner electricity and depend less on imported fuel
 - C. stop the use of all machines
 - D. increase the smoke in the cities
- 37.** A pupil who wishes to become an engineer should pay particular attention to
- A. science and mathematics
 - B. handwriting only
 - C. games and sports only
 - D. memorising long poems
- 38.** The BEST way for a Basic 7 pupil to prepare for a career in science is to
- A. take part in practical work, read widely and ask questions
 - B. wait until senior high school before studying science
 - C. avoid difficult topics
 - D. copy answers from friends
- 39.** A school science club wishing to learn about careers in science would benefit MOST from
- A. reading only the names of scientists
 - B. collecting money from pupils
 - C. holding a football competition each week
 - D. inviting scientists and visiting laboratories, farms and factories
- 40.** The MAIN purpose of building a portfolio on the impact of science and technology is to
- A. gather and organise evidence of what has been found out, for presentation
 - B. keep the findings away from other pupils
 - C. replace the class textbook
 - D. avoid doing any research

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 17: Solar Sys

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The solar system is made up of
 - A. the Earth and the Moon only
 - B. all the stars in the sky
 - C. the clouds and the air above the Earth
 - D. the Sun and all the bodies that move round it
2. At the centre of the solar system is
 - A. the Sun
 - B. Mercury
 - C. the Earth
 - D. the Moon
3. The Sun is best described as
 - A. a moon
 - B. a star
 - C. a cloud of water
 - D. a planet
4. The Sun is important to the solar system because it
 - A. moves round the Earth each day
 - B. supplies the planets with water
 - C. reflects light from the planets
 - D. supplies light and heat and holds the planets in their paths
5. A planet differs from a star in that a planet
 - A. gives out its own light and heat
 - B. is always nearer to the Earth than the Moon
 - C. is always larger than a star
 - D. does not give out its own light but shines by reflecting sunlight
6. The force that keeps the planets moving round the Sun instead of flying away is
 - A. the heat of the Sun
 - B. the magnetism of the Earth
 - C. the gravitational pull of the Sun
 - D. the push of the wind in space

- 7.** Which of the following is NOT a member of the solar system?
- A. The Moon
 - B. The Milky Way
 - C. Mars
 - D. The Sun
- 8.** The planets nearest to the Sun are called
- A. the outer planets
 - B. the stars
 - C. the satellites
 - D. the inner planets
- 9.** How many inner planets are there in the solar system?
- A. Six
 - B. Four
 - C. Two
 - D. Eight
- 10.** The inner planets, in order from the Sun, are
- A. Venus, Mercury, Mars, Earth
 - B. Mercury, Venus, Earth, Mars
 - C. Mercury, Earth, Venus, Mars
 - D. Earth, Mars, Mercury, Venus
- 11.** The inner planets are also called the rocky planets because they
- A. are covered with ice
 - B. are made mainly of gas
 - C. have solid, rocky surfaces
 - D. are made of burning rock
- 12.** The planet nearest to the Sun is
- A. Earth
 - B. Mars
 - C. Venus
 - D. Mercury
- 13.** The fourth planet from the Sun is
- A. Earth
 - B. Venus
 - C. Mercury
 - D. Mars

- 14.** Which inner planet is known to support life?
- A. Earth
 - B. Mars
 - C. Venus
 - D. Mercury
- 15.** Mars is often called the red planet because
- A. it is very close to the Sun
 - B. it is the hottest planet
 - C. the rocks and dust on its surface are reddish in colour
 - D. it gives out red light of its own
- 16.** Which of the following is a property shared by all the inner planets?
- A. They all have many moons
 - B. They all have rings around them
 - C. They are all small and have solid surfaces
 - D. They are all covered with liquid water
- 17.** A very large group of stars, dust and gas held together by gravity is called
- A. a planet
 - B. a solar system
 - C. a satellite
 - D. a galaxy
- 18.** The galaxy in which our solar system is found is called
- A. the Milky Way
 - B. the Great Bear
 - C. the Orion Belt
 - D. the Southern Cross
- 19.** On a clear, dark night away from town lights, the Milky Way appears as
- A. a faint band of light stretching across the sky
 - B. a single very bright star
 - C. a line of moving lights
 - D. a round red object
- 20.** Which of the following statements about the Milky Way is correct?
- A. The Milky Way is the path followed by the Earth
 - B. The Milky Way is found inside the solar system
 - C. The solar system is only a very small part of the Milky Way
 - D. The Milky Way is another name for the Sun

- 21.** Arrange the following from the smallest to the largest: galaxy, planet, solar system.
- A. Planet, galaxy, solar system
 - B. Planet, solar system, galaxy
 - C. Galaxy, solar system, planet
 - D. Solar system, planet, galaxy
- 22.** Street lights make it difficult to see the Milky Way from a town in Ghana because
- A. tall buildings block the light of the stars
 - B. the Milky Way moves away from towns
 - C. the bright lights of the town drown the faint light from the stars
 - D. the Milky Way can be seen only in the rainy season
- 23.** The shape of the path along which a planet moves round the Sun is
- A. elliptical (oval)
 - B. a perfect circle
 - C. a square
 - D. a straight line
- 24.** The path along which a planet travels round the Sun is called its
- A. axis
 - B. orbit
 - C. equator
 - D. satellite
- 25.** The turning of a planet on its own axis is called
- A. rotation
 - B. radiation
 - C. revolution
 - D. reflection
- 26.** The movement of a planet round the Sun is called
- A. orbit tilting
 - B. rotation
 - C. vibration
 - D. revolution
- 27.** The rotation of the Earth on its axis causes
- A. the seasons of the year
 - B. the tides of the sea
 - C. day and night
 - D. the phases of the Moon

- 28.** A planet that is far from the Sun takes a longer time to go round the Sun than one that is near. This is mainly because the far planet
- A. has a much longer path to travel
 - B. stops from time to time on the way
 - C. is pulled backwards by the stars
 - D. is heavier than the near planet
- 29.** One complete revolution of the Earth round the Sun takes about
- A. 24 hours
 - B. 365 days
 - C. 88 days
 - D. 30 days
- 30.** Which of the following is a property of Mercury?
- A. It is the smallest of the inner planets and has almost no atmosphere
 - B. It has a thick atmosphere of oxygen
 - C. It is covered with large oceans of water
 - D. It is the largest planet in the solar system
- 31.** Mercury has a very hot day side and a very cold night side mainly because it
- A. is far from the Sun
 - B. turns very fast on its axis
 - C. is covered with white ice
 - D. has almost no atmosphere to hold in the heat
- 32.** Mercury takes about 88 Earth days to go round the Sun once. This means that Mercury
- A. does not move round the Sun at all
 - B. has a much shorter year than the Earth
 - C. has a much longer year than the Earth
 - D. has a year of the same length as that of the Earth
- 33.** Venus is the hottest planet in the solar system because
- A. it is the planet nearest to the Sun
 - B. its thick atmosphere traps the heat from the Sun
 - C. it gives out heat of its own
 - D. it is made of burning coal
- 34.** Venus is often called the morning star or the evening star because it
- A. appears only once each year
 - B. is the brightest star in the sky
 - C. can be seen shining brightly near sunrise or sunset
 - D. gives out its own light at dawn

- 35.** Which statement about the sizes of Venus and the Earth is correct?
- A. Venus is nearly the same size as the Earth
 - B. Venus is the largest planet in the solar system
 - C. Venus is many times larger than the Earth
 - D. Venus is smaller than the Moon
- 36.** Mercury and Venus are alike in that both
- A. have surfaces covered with water
 - B. have many moons
 - C. are larger than the Earth
 - D. move round the Sun in paths inside that of the Earth
- 37.** A model of the solar system is useful mainly because it
- A. helps us to see the arrangement and movement of bodies that are too far away to visit
 - B. can be used to measure the heat of the Sun
 - C. shows the true sizes and distances of the planets
 - D. makes the planets move faster
- 38.** In building a model of the solar system from balls of different sizes, the largest ball should stand for
- A. Venus
 - B. the Earth
 - C. the Sun
 - D. Mercury
- 39.** A pupil at Koforidua made a model in which the four inner planets were placed in a straight line beside the Sun. The main fault in this model is that
- A. the planets should be bigger than the Sun
 - B. the Sun should be placed at the edge of the model
 - C. the planets move round the Sun along separate paths and are not in a line
 - D. the planets should all be the same size
- 40.** Which material would be the MOST suitable for showing the path of a planet in a model of the solar system?
- A. A square piece of cardboard
 - B. A loop of wire bent into an oval shape
 - C. A straight stick
 - D. A short piece of rope tied in a knot

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 18: Understanding the

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A landform is BEST described as
 - A. a map drawn by a surveyor
 - B. a kind of soil used for farming
 - C. a natural feature of the surface of the Earth, such as a mountain or a plain
 - D. a building put up by people
2. A wide stretch of flat, low land is called a
 - A. plateau
 - B. valley
 - C. mountain
 - D. plain
3. A piece of high land with a broad, flat top is a
 - A. plateau
 - B. valley
 - C. delta
 - D. plain
4. The low land lying between two hills or mountains is a
 - A. plateau
 - B. plain
 - C. peak
 - D. valley
5. The highest mountain in Ghana is
 - A. Mount Afadja (Afadjato)
 - B. Mount Kenya
 - C. Mount Kilimanjaro
 - D. Mount Cameroon

- 6.** The Accra Plains and the Afram Plains are examples of the landform called the
- A. plateau
 - B. plain
 - C. valley
 - D. mountain
- 7.** Which group of animals would one expect to find MOST commonly on a grassy plain?
- A. bats, baboons and rock hyraxes
 - B. grasscutters, antelopes and cattle
 - C. crabs, oysters and starfish
 - D. tilapia, catfish and mudfish
- 8.** Mangrove plants and crabs are found MAINLY in
- A. the roof of a cave
 - B. the middle of a desert
 - C. dry rocky hill tops
 - D. muddy ground along the coast and river mouths
- 9.** The special features that help a living thing to survive where it lives are called
- A. habitats
 - B. associations
 - C. adaptations
 - D. competitions
- 10.** The thick fur of an animal living on a cold mountain helps it to
- A. keep its body warm
 - B. store water in its body
 - C. find its food in the dark
 - D. run faster on the rocks
- 11.** Mountain goats and rock hyraxes can move about on steep rocky slopes because they have
- A. webbed feet for swimming
 - B. strong legs and hooves or pads that grip the rock
 - C. wide flat feet for walking on mud
 - D. long tails for hanging from trees
- 12.** The long legs of the egret that feeds in a marshy plain help it to
- A. dig deep holes in dry soil
 - B. climb tall trees quickly
 - C. wade through shallow water without wetting its body
 - D. fly faster than other birds

- 13.** An antelope living on an open plain escapes from its enemies MAINLY by
- A. swimming across the plain
 - B. burrowing into the rock
 - C. changing into another animal
 - D. running very fast on its long slender legs
- 14.** A fish living in a fast-flowing stream in a valley is helped to stay in place by its
- A. streamlined body and strong fins
 - B. hard shell and eight legs
 - C. hollow bones and feathers
 - D. thick fur and long ears
- 15.** The sandy colour of many animals of the dry northern plains is useful because it
- A. makes them attractive to hunters
 - B. helps them to dig faster
 - C. helps them to blend with their surroundings and escape notice
 - D. keeps insects away from them
- 16.** Burrowing into the ground during the hottest part of the day helps a small animal of the plains to
- A. grow a longer tail
 - B. find more sunlight
 - C. increase the loss of water from its body
 - D. avoid the heat and reduce the loss of water from its body
- 17.** Plants growing in a dry, open plain often have long roots in order to
- A. catch insects for food
 - B. hold leaves above the clouds
 - C. produce more flowers in the dry season
 - D. reach water deep in the soil
- 18.** The small, needle-like leaves of some plants in dry places help them to
- A. lose more water by transpiration
 - B. lose less water by transpiration
 - C. trap more rain water
 - D. store more seeds
- 19.** The thick fleshy stem of a cactus-like plant is useful for
- A. killing insects
 - B. storing water
 - C. attracting rainfall
 - D. storing oxygen

- 20.** Mangrove plants growing in soft muddy soil are supported by
- A. a single straight tap root only
 - B. woody stems that lie flat in the mud
 - C. floating leaves
 - D. stilt (prop) roots that spread out above the mud
- 21.** Trees of the thick forest in a wet valley are usually
- A. short, with needle-like leaves
 - B. tall, with broad leaves that catch the sunlight
 - C. leafless throughout the year
 - D. grown only in sandy soil
- 22.** Grasses survive the yearly bush fires of the savanna plains MAINLY because they
- A. grow taller than the flames
 - B. are too wet to burn at all
 - C. have underground parts that sprout again after the fire
 - D. are protected by their flowers
- 23.** Some plants on rocky hillsides grow close to the ground. This helps them to
- A. lose more water quickly
 - B. resist being blown away by strong winds
 - C. receive less sunlight
 - D. escape from grazing animals completely
- 24.** A plant that grows on a plateau with shallow, stony soil is likely to have
- A. roots that spread out widely near the surface
 - B. no roots at all
 - C. roots that float in water
 - D. one very deep root only
- 25.** An association in which BOTH living things benefit is called
- A. predation
 - B. parasitism
 - C. commensalism
 - D. mutualism
- 26.** In parasitism, the parasite
- A. is harmed while the host benefits
 - B. and the host are both harmed
 - C. benefits while the host is harmed
 - D. and the host both benefit equally

- 27.** An association in which one living thing benefits while the other is neither helped nor harmed is
- A. mutualism
 - B. parasitism
 - C. competition
 - D. commensalism
- 28.** A bee visiting a flower for nectar and carrying pollen to the next flower is an example of
- A. predation
 - B. mutualism
 - C. parasitism
 - D. commensalism
- 29.** A tapeworm living in the intestine of a goat shows
- A. mutualism
 - B. competition
 - C. commensalism
 - D. parasitism
- 30.** A cattle egret that feeds on the insects disturbed by a grazing cow, without affecting the cow, shows
- A. mutualism
 - B. commensalism
 - C. parasitism
 - D. predation
- 31.** Which of the following is a parasite of a flowering plant?
- A. an earthworm in the soil below the tree
 - B. a bee resting on a flower
 - C. mistletoe growing on the branch of a tree
 - D. a bird building a nest in the branches
- 32.** Root nodule bacteria living in the roots of a groundnut plant supply the plant with nitrogen and receive food from it. This is
- A. commensalism
 - B. a food chain
 - C. mutualism
 - D. parasitism
- 33.** A lion hunting and killing an antelope is BEST described as
- A. predation
 - B. mutualism
 - C. commensalism
 - D. parasitism

- 34.** The place where a plant or an animal naturally lives is its
- A. habitat
 - B. population
 - C. adaptation
 - D. association
- 35.** Two goats feeding on the same small patch of grass are said to be in
- A. parasitism
 - B. competition
 - C. mutualism
 - D. commensalism
- 36.** A chameleon changing its colour to match the branch it rests on is an example of
- A. camouflage
 - B. pollination
 - C. hibernation
 - D. migration
- 37.** Birds flying from one region to another when the dry season comes are showing
- A. migration
 - B. camouflage
 - C. germination
 - D. parasitism
- 38.** A pupil studying the animals of a nearby valley should FIRST
- A. copy the names of animals from a friend
 - B. write the report before the visit
 - C. visit the valley and observe and record the animals seen there
 - D. collect money from the class
- 39.** Which of the following is the BEST record to keep in an album of the plants and animals of a landform?
- A. a labelled picture or drawing of each one, with the place and date it was seen
 - B. the price of each animal in the market
 - C. a list of names only, without any detail
 - D. the names of the pupils in the class only
- 40.** If the trees of a valley are all cut down, the animals living there will MOST likely
- A. change into plants
 - B. grow larger and multiply quickly
 - C. lose their shelter and food and move away or die
 - D. remain exactly as they were

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 19: Waste Management

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Waste is BEST described as
 - A. any material that is sold in the market
 - B. any material that is buried in the soil
 - C. any material that is no longer wanted by the person who produced it
 - D. only the rubbish swept from a room
2. Peels, leftover food and dry leaves are examples of
 - A. metallic waste
 - B. electronic waste
 - C. organic (biodegradable) waste
 - D. chemical waste
3. Water sachets, plastic bottles and rubber bags are examples of
 - A. organic waste
 - B. metallic waste
 - C. glass waste
 - D. plastic waste
4. Empty milk tins and old roofing sheets are examples of
 - A. liquid waste
 - B. paper waste
 - C. plastic waste
 - D. metallic waste
5. Used needles, soiled bandages and expired drugs from a clinic are classified as
 - A. paper waste
 - B. domestic waste
 - C. medical waste
 - D. agricultural waste
6. Dirty water from a bathroom and a kitchen is an example of
 - A. gaseous waste
 - B. toxic metallic waste
 - C. solid waste
 - D. liquid waste

- 7.** Smoke from a lorry and fumes from a factory chimney are examples of
- A. gaseous waste
 - B. liquid waste
 - C. solid waste
 - D. organic waste
- 8.** Spoilt mobile phones, old radios and broken television sets are known as
- A. medical waste
 - B. electronic waste
 - C. agricultural waste
 - D. paper waste
- 9.** Biodegradable waste is waste that
- A. cannot be broken down at all
 - B. can be burnt in an open fire
 - C. can be broken down by micro-organisms into simple substances
 - D. can be sold to a recycling factory
- 10.** Which of the following wastes is NON-biodegradable?
- A. A sheet of old newspaper
 - B. A plastic water sachet
 - C. A banana peel
 - D. Maize husks
- 11.** The organisms chiefly responsible for breaking down biodegradable waste are
- A. birds and bats
 - B. bacteria and fungi
 - C. fish and frogs
 - D. grasshoppers and bees
- 12.** A plastic bottle buried in the soil is still whole after twenty years because plastic
- A. absorbs water from the soil
 - B. is a living material
 - C. is heavier than soil
 - D. is not broken down by micro-organisms
- 13.** Waste that rots and gives off a bad smell is MOST likely to be
- A. food remains and other organic waste
 - B. broken glass
 - C. aluminium cans
 - D. pieces of rubber tyre

- 14.** Heaps of rotting refuse near homes are dangerous mainly because they
- A. attract rain to the area
 - B. lower the temperature of the air
 - C. make the soil too rich for crops
 - D. breed houseflies, mosquitoes and rats that spread disease
- 15.** Choked gutters filled with plastic waste in a town lead directly to
- A. an increase in soil fertility
 - B. flooding when it rains
 - C. a fall in the price of plastics
 - D. cleaner drinking water
- 16.** Burning rubber and plastic waste in the open is discouraged because it
- A. cools the surrounding air
 - B. destroys the waste completely with no effect
 - C. releases poisonous gases and smoke into the air
 - D. makes the plastics biodegradable
- 17.** The three Rs of good waste management are
- A. remove, rinse and rest
 - B. reduce, reuse and recycle
 - C. read, record and report
 - D. repair, rebuild and resell
- 18.** Carrying a basket to the market instead of collecting several plastic bags is an example of
- A. recycling waste
 - B. burying waste
 - C. reducing waste
 - D. burning waste
- 19.** Using an empty tin as a container for seedlings is an example of
- A. recycling waste
 - B. reducing waste
 - C. dumping waste
 - D. reusing waste
- 20.** Collecting plastic bottles and sending them to a factory where they are melted and made into new articles is
- A. recycling
 - B. incineration
 - C. composting
 - D. reusing

- 21.** Separating waste into different containers at the point where it is produced is called
- A. waste recycling
 - B. waste burning
 - C. waste collection
 - D. waste segregation (sorting)
- 22.** The MAIN advantage of sorting waste at home is that it
- A. makes it easier to compost, reuse or recycle each kind of waste
 - B. removes the need for dustbins
 - C. makes the waste disappear
 - D. increases the total amount of waste
- 23.** Household refuse kept in a covered container is better than refuse left in an open pan because the cover
- A. keeps out flies and prevents bad smells from spreading
 - B. makes the refuse rot faster
 - C. turns the refuse into compost at once
 - D. adds weight to the refuse
- 24.** Treating waste in a properly designed incinerator is better than open burning because the incinerator
- A. uses no fuel at all
 - B. burns the waste at a high temperature and controls the gases released
 - C. turns all the waste into drinking water
 - D. produces no ash
- 25.** Which of the following is the BEST method of disposing of broken glass at home?
- A. Wrapping it securely and placing it in a bin for hard waste
 - B. Putting it loose into a sack of food waste
 - C. Burying it in the vegetable bed
 - D. Throwing it into the gutter
- 26.** Composting is the process by which
- A. waste water is boiled to make it pure
 - B. plastic waste is melted into new bottles
 - C. organic waste is allowed to decay to form manure
 - D. metal waste is beaten into new tools
- 27.** The materials MOST suitable for making compost are
- A. old batteries and spoilt phones
 - B. grass cuttings, peels, dry leaves and animal droppings
 - C. tins and broken glass
 - D. water sachets and plastic bottles

- 28.** Water is sprinkled on a compost heap in order to
- A. wash away the bad smell
 - B. cool the heap and stop decay
 - C. keep it moist so that the micro-organisms can act
 - D. make the heap heavier
- 29.** A compost heap is turned over from time to time mainly to
- A. allow air to reach all parts of the heap
 - B. remove the manure formed
 - C. make the heap look tidy
 - D. stop the micro-organisms from working
- 30.** The warmth felt inside a compost heap is evidence that
- A. the Sun has heated the whole heap
 - B. micro-organisms are active and respiration is taking place
 - C. water is freezing inside the heap
 - D. the waste is burning with a flame
- 31.** Compost added to garden soil is useful because it
- A. kills all the organisms in the soil
 - B. turns the soil into sand
 - C. supplies plant nutrients and improves the structure of the soil
 - D. makes the soil hold no water
- 32.** The use of a compost pit to treat kitchen waste shows that waste can be
- A. changed into plastic
 - B. stored for ever without change
 - C. made to disappear completely
 - D. turned into a useful resource
- 33.** Dumping refuse into a stream used by a village is dangerous because it
- A. pollutes the water and can cause diseases such as cholera
 - B. increases the number of fish
 - C. purifies the water naturally
 - D. makes the stream flow faster
- 34.** A common disease spread by houseflies that breed on uncovered refuse is
- A. sickle-cell disease
 - B. typhoid fever
 - C. night blindness
 - D. ringworm

- 35.** Mosquitoes that spread malaria breed MOST easily in
- A. water collected in discarded tins and old tyres
 - B. dry heaps of sand
 - C. burning rubbish
 - D. swept and clean compounds
- 36.** Before carrying out a clean-up project in the community, pupils should FIRST
- A. research and plan the work and obtain permission from the authorities
 - B. begin by burning all the waste they find
 - C. clean only their own homes
 - D. collect money from passers-by
- 37.** The MOST suitable protective items for pupils taking part in a clean-up exercise are
- A. sunglasses and wrist watches
 - B. hand gloves, nose masks and boots
 - C. school bags and books
 - D. sandals and short sleeves
- 38.** At the end of a waste management project, the pupils should
- A. keep the findings a secret
 - B. forget about the project at once
 - C. repeat exactly the same work the next day
 - D. write and present a report on what they did and found
- 39.** The BEST way for a school to keep its compound permanently clean is to
- A. burn all waste at the back of the classrooms
 - B. ask pupils to carry their waste home every day
 - C. provide labelled bins and agree on a routine for emptying and sorting them
 - D. sweep only when visitors are expected
- 40.** A community that practises good waste management will MOST likely experience
- A. more flooding in the rainy season
 - B. an increase in malaria cases
 - C. a greater number of refuse heaps
 - D. fewer diseases and a cleaner environment

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 20: Water Cycle

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Which of the following is a natural source of water?
 - A. A water tanker
 - B. A sachet of water
 - C. A tap
 - D. A river
2. The largest natural store of water on the earth is
 - A. the sea
 - B. the stream
 - C. the well
 - D. the lagoon
3. Water that collects under the ground and is reached by digging a well is called
 - A. rain water
 - B. sea water
 - C. ground water
 - D. pipe water
4. Lake Bosomtwe, the Volta Lake and the Pra River are all examples of
 - A. underground streams
 - B. sources of sea water
 - C. man-made reservoirs only
 - D. surface water bodies
5. Which of the following best explains why the water in the sea cannot be drunk straight away?
 - A. It contains a large amount of dissolved salt
 - B. It never evaporates
 - C. It is always too cold
 - D. It contains no dissolved substances

- 6.** The change of water from the liquid state into water vapour is called
- A. precipitation
 - B. evaporation
 - C. condensation
 - D. melting
- 7.** The main source of the energy that drives the water cycle is
- A. the moon
 - B. the wind
 - C. the soil
 - D. the sun
- 8.** Clothes spread on a line in the open dry quickly because the water in them
- A. condenses on the cloth
 - B. freezes in the air
 - C. evaporates into the air
 - D. sinks into the ground
- 9.** The loss of water vapour from the leaves of plants to the atmosphere is called
- A. respiration
 - B. transpiration
 - C. condensation
 - D. germination
- 10.** Which of the following conditions will make water in an open pan evaporate fastest?
- A. A shaded, humid evening
 - B. A cool, still night
 - C. A hot, windy afternoon
 - D. A cold, rainy morning
- 11.** Two equal volumes of water were poured into a wide basin and a narrow bottle and left in the sun. The water in the basin disappeared faster because
- A. it had a larger surface exposed to the air
 - B. it was heavier than the water in the bottle
 - C. it was colder than the water in the bottle
 - D. it contained more dissolved air
- 12.** During evaporation from the sea, the salt in the sea water
- A. is left behind in the sea
 - B. falls back as salty rain
 - C. rises with the water vapour
 - D. changes into water vapour

- 13.** A farmer at Techiman covered the soil around his young plants with dry grass. This practice mainly reduces
- A. condensation in the air
 - B. transpiration from the leaves
 - C. precipitation over the farm
 - D. evaporation of water from the soil
- 14.** The change of water vapour into tiny drops of liquid water is called
- A. evaporation
 - B. transpiration
 - C. precipitation
 - D. condensation
- 15.** Water vapour rising into the atmosphere condenses mainly because the air above
- A. becomes warmer with height
 - B. moves faster with height
 - C. becomes cooler with height
 - D. contains no dust
- 16.** Clouds are made up of
- A. pure oxygen gas
 - B. tiny drops of water and ice crystals
 - C. smoke from factories
 - D. dry dust from the desert
- 17.** Drops of water form on the outside of a cold bottle of mineral water left on a table because
- A. the glass melts a little
 - B. water leaks out through the glass
 - C. the water inside evaporates through the glass
 - D. water vapour in the air condenses on the cold surface
- 18.** Dew that is seen on grass early in the morning is formed by
- A. condensation of water vapour on the cool grass
 - B. light rain that falls at dawn
 - C. evaporation of water from the soil
 - D. transpiration by the grass at night
- 19.** In the water cycle, condensation takes place
- A. on the surface of the sea
 - B. high in the atmosphere where the air is cool
 - C. inside the roots of plants
 - D. deep inside the soil

- 20.** Which process is the direct opposite of evaporation?
- A. Condensation
 - B. Precipitation
 - C. Infiltration
 - D. Transpiration
- 21.** Water that falls from clouds to the earth as rain, hail or snow is known as
- A. transpiration
 - B. precipitation
 - C. evaporation
 - D. condensation
- 22.** Rain falls from a cloud when the water drops in the cloud
- A. join together and become too heavy to stay in the air
 - B. become warmer than the air around them
 - C. change back into water vapour
 - D. are pushed up by the wind
- 23.** The commonest form of precipitation in Ghana is
- A. hail
 - B. rain
 - C. sleet
 - D. snow
- 24.** Rain water that flows over the surface of the land into streams and rivers is called
- A. condensation
 - B. transpiration
 - C. run-off
 - D. evaporation
- 25.** The movement of rain water downwards into the soil is called
- A. evaporation
 - B. transpiration
 - C. infiltration
 - D. precipitation
- 26.** After a heavy downpour at Kumasi, most of the rain water that fell on the tarred roads
- A. soaked into the tar
 - B. was taken up by the roads as ground water
 - C. evaporated immediately
 - D. ran off quickly into gutters and streams

- 27.** Flooding is common in towns where many open spaces have been covered with concrete, mainly because
- A. less water soaks into the ground, so run-off increases
 - B. plants take up less water from the air
 - C. more water evaporates from the concrete
 - D. concrete causes heavier rainfall
- 28.** The correct order of the stages of the water cycle is
- A. collection, precipitation, condensation, evaporation
 - B. evaporation, condensation, precipitation, collection
 - C. condensation, evaporation, collection, precipitation
 - D. precipitation, evaporation, condensation, collection
- 29.** The water cycle is described as a repeated pattern of change in nature because
- A. it happens only in the rainy season
 - B. it takes place once in every year
 - C. new water is made each time it rains
 - D. the same stages follow one another again and again without ending
- 30.** The total amount of water on the earth
- A. decreases each time the sun shines
 - B. remains about the same, as the water is used over and over again
 - C. increases each time it rains
 - D. is made fresh by plants each year
- 31.** Which of the following returns water from the land back to the sea, completing the cycle?
- A. Rivers and streams
 - B. Wind
 - C. Dew
 - D. Clouds
- 32.** A pupil drew a flow chart of the water cycle but ended it at precipitation. The chart is incomplete because it fails to show that
- A. clouds are made of water drops
 - B. rain water is always clean
 - C. the water returns to the sources from which it evaporates again
 - D. plants also lose water
- 33.** The water that a pupil drinks today may once have been part of a river that flowed many years ago. This is possible because
- A. fresh water is created by rainfall
 - B. water does not change state
 - C. water is used again and again in the water cycle

D. rivers never dry up

34. The water cycle is important to farmers mainly because it

- A. removes weeds from the farm
- B. provides rain for the growth of crops
- C. makes the soil harder
- D. provides salt for the soil

35. As water vapour condenses in the atmosphere, energy is released. This released energy helps to

- A. form salt in the air
- B. increase run-off on land
- C. warm the surrounding environment
- D. cool the sea water

36. The water cycle raises the water table of an area by

- A. allowing rain water to soak into the ground and collect below the surface
- B. keeping the soil surface dry
- C. carrying water away in rivers
- D. increasing evaporation from the sea

37. Water moving through the cycle is described as a carrier of nutrients because it

- A. stores nutrients only in clouds
- B. destroys the nutrients it meets
- C. turns mineral salts into gas
- D. dissolves mineral salts and carries them to soils, rivers and living things

38. Which of the following best explains how the water cycle helps to provide clean water for a community?

- A. Water is boiled in the clouds before it falls
- B. Water that evaporates leaves its dissolved salts and dirt behind and returns as fairly pure rain
- C. Clouds filter out the water and keep the clean part
- D. Rain water collects salt from the air as it falls

39. A town has cut down the trees on the hills around it and the rains have become less reliable. The most likely reason is that

- A. the trees used to block the clouds
- B. the soil now holds too much water
- C. trees prevent water from evaporating from rivers
- D. with fewer trees there is less transpiration, so less water vapour enters the air

- 40.** A community wants to reduce the flooding that follows every heavy rain while also raising its water table. The BEST action to take is to
- A. pump the flood water into the nearest stream each time
 - B. build higher walls round the houses
 - C. plant trees and grasses and keep open ground so that rain water soaks in
 - D. pave all the open ground with concrete

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.